1 Introduction

This dissertation focuses on the task of remote sensing multi-label scene classification on MLRSNet [1]**.** Since MLRSNet simultaneously provides primary scene labels and secondary semantic labels, this dissertation adopts the following two-stage decomposition:

$$p(y_{pri}, y_{sec} \mid x) = p(y_{pri} \mid x) p(y_{sec} \mid x, y_{pri}). \tag{1.1}$$

The first stage predicts the primary category, and the second stage predicts the secondary labels under a given primary condition. The basis for this decomposition is that the secondary label space is sparse and long-tail imbalanced, there are statistical dependencies among labels, and the primary label significantly changes the conditional distribution of the secondary labels [1][6][9][4].

1.1 Background and Context

Remote sensing images are widely used in scenarios such as land-use analysis, environmental monitoring, infrastructure inspection, and disaster response. In high-resolution aerial remote sensing images, multiple semantic elements often appear simultaneously; therefore, single-label scene classification cannot satisfy many practical application needs [1][2]. MLRSNet is suitable for the problem setting in this dissertation because it simultaneously provides scene-level labels and multi-label semantic annotations within the same dataset [1].

1.2 Scope and Objectives

The scope of this dissertation is limited to RGB images in MLRSNet.

This project has five main objectives:

- Analyse the structure of the dataset, including class balance, label sparsity, and label dependency relationships.
- Construct a two-stage framework for primary scene classification and conditional secondary label prediction.
- Design an improved ASL-GB on the basis of fixed-exponent ASL and analyse its mechanism in long-tail negative-class reweighting and conditional multi-label scenes.
- Compare the differences between the independent baseline head and correlation-aware, imbalance-aware variants under the same conditions.
- Use standard multi-label metrics and diagnostic analyses to evaluate the sources of model performance and cascade errors.

In terms of the research route, the dataset analysis and dataset split are completed first, then a two-stage model based on DINOv3 is constructed; next, the improved ASL-GB is introduced on the basis of fixed-exponent ASL and the properties of its negative branch are analysed; finally, the framework is comprehensively evaluated by combining overall metrics, label-level results, and cascade error diagnosis.

1.3 Achievements

This dissertation has completed four items of work:

First, a systematic statistical analysis was conducted on the class distribution, primary-secondary relationships, co-occurrence structure, and conditional dependence in MLRSNet;

Second, a two-stage framework based on DINOv3 was constructed, and the way in which the primary prior enters the second stage was formally described.

Third, ASL-GB was proposed. While retaining negative-side truncation, it introduces a training-step-dependent negative-side focusing exponent, and its influence on the reweighting of active negative coordinates is analysed.

Fourth, through same-backbone comparison, label-level analysis, and Oracle/Cascade comparison, the contributions of conditional correlation modelling and loss design are separated, and structured cascade error is identified as the current main limitation.

Under the same backbone and training protocol, the proposed two-stage conditional framework achieves better overall performance than the independent secondary baseline, with the most consistent gains observed in ranking-oriented metrics and label-structure consistency. These results indicate that explicit primary-conditioned secondary modelling is effective for MLRSNet multi-label scene understanding; the primary source of gain is conditional decomposition, while ASL-GB provides additional gains under long-tail conditions.

1.4 Overview of Dissertation

Chapter 2 reviews related research on remote sensing scene classification, multi-label learning, Transformer backbones, and the limitations of independent label modelling. Chapter 3 introduces the dataset audit results and the process of preparing experimental data. Chapter 4 presents the design of the two-stage model and an analysis of its modelling rationality. Chapter 5 introduces the loss design of the improved ASL-GB and focuses on the comparative-static properties of its negative branch. Chapter 6 reports the experimental results, error analysis, and discussion. Chapter 7 summarises the main findings of the dissertation, its research limitations, and future directions.

2 State-of-The-Art

2.1 Early Approaches

Early studies on remote sensing scene classification primarily adopted hand-crafted methods, in which feature extraction (e.g., color histograms, texture descriptors, and SIFT variants) and classification were designed and optimized separately [2][3]. The bag-of-visual-words paradigm improved reproducibility on small and medium-scale benchmarks; however, its order-invariant local-statistics representation had limited capacity to characterize long-range spatial dependence and complex semantic composition [3].

In parallel, early research in multi-label learning had already shown that the label-independence assumption is frequently violated in practical tasks [4][5]. Nevertheless, many remote sensing implementations continued to treat labels as independent sigmoid outputs because this formulation is simple and computationally stable. Under this setting, the model is optimized for per-label marginal fitting, whereas evaluation often depends on sample-level consistency and the correct characterization of label co-occurrence patterns [4]. This discrepancy indicates a structural inconsistency between modelling assumptions and the underlying annotation process.

Table 1 summarises the reasons why earlier approaches remain important baselines but are insufficient for the target problem addressed in this dissertation.

Table 1. Limitations of earlier paradigms.

Earlier paradigm	Representative references	Main contribution	Structural limitation for this dissertation
Hand-crafted features + shallow classifiers	[2][3]	Established benchmark protocols and reproducible pipelines	Limited ability to jointly represent high-level context and cross-object composition in complex scenes
Independent one-vs-rest multi-label outputs	[4][5]	Simple optimization procedure and clear per-label probabilistic interpretation	Neglects conditional dependence and scene-specific co-occurrence, which is inconsistent with the annotation structure of MLRSNet [1]
Small closed-set scene benchmarks	[2]	Enabled controlled comparison in early studies	Provides limited support for analysing hierarchical annotation and long-tail multi-label dependence

2.2 Dependency-Aware Methods

The first group of advanced studies focuses on explicit dependency modelling among labels. Relation Network methods learn pairwise interactions and improve aerial multi-label recognition relative to independent prediction heads [6]. Latent semantic dependency methods further introduce hidden dependency structures, thereby improving robustness when direct co-occurrence information is sparse or noisy [7]. CNN+GNN hybrid models represent labels as graph nodes and use message passing to propagate relational evidence [8]. More recent Transformer-based label-interaction models employ attention mechanisms to couple global visual tokens with semantic label tokens, thereby improving representational flexibility [9].

These studies indicate that dependency modelling is an important component of multi-label remote sensing [6][9]. However, most of them still formulate prediction as an unconditional mapping from an image to all labels. For MLRSNet, in which a primary scene label and secondary semantic labels are jointly annotated, such a formulation does not make full use of condition-specific information associated with the primary scene variable [1].

Table 2 presents a comparative assessment of this first group of methods.

Table 2. Comparison of dependency-modelling methods.

Method family	Representative paper	Dependency type captured	Practical strength	Limitation relative to this study
Relation network	Hua et al. [6]	Primarily pairwise relations	Simple integration with CNN backbones	Pairwise structure alone may be insufficient to capture higher-order scene-conditioned patterns
Latent semantic dependency	Ji et al. [7]	Implicit higher-order semantics	Improved robustness under sparse co-occurrence	Dependency semantics are less directly controllable and more difficult to diagnose
CNN + GNN	Li et al. [8]	Graph-based message passing among labels	Clear relational inductive bias	The formulation is usually unconditional with respect to an explicit scene variable
Semantic Transformer dependency modeling	Wu et al. [9]	Attention-based global interactions	Strong integration of global contextual information	Performance gains are often associated with model capacity, but the framework does not explicitly decompose uncertainty from primary scenes to secondary labels

2.3 Representation and Loss Advances

The second group improves performance through representation scale, pre-training strategy, and loss design. Vision Transformers and self-supervised pre-training improve global-context modelling and transfer efficiency [10][11], while multi-scale Transformer architectures improve adaptability to variation in object size [12]. In parallel, studies on loss design, including the Focal Loss and Asymmetric Loss families, address class imbalance and the dominance of hard negatives [14][13][15]. More general studies on imbalanced multi-label learning further show that data stratification and label-frequency-aware evaluation have a substantial influence on empirical conclusions [16][18].

This line of research primarily addresses optimization and representation issues under long-tail label distributions, including representational capacity and mitigation of gradient bias. However, two limitations remain in relation to the present dissertation. First, stronger backbones do not automatically encode hierarchical annotation semantics unless the task formulation explicitly exposes such structure. Second, improved losses can rebalance positive and negative gradients, but they do not by themselves correct an inappropriate conditional factorization.

Table 3 summarises these observations.

Table 3. Comparison of representation and loss-design approaches.

Technical line	Representative references	Problem addressed	What remains unresolved for this dissertation
ViT and self-supervised pre-training	[10][11]	Long-range contextual modelling and transfer learning	Does not in itself impose a primary-to-secondary conditional decomposition
Multi-scale Transformer design	[12]	Multi-scale spatial heterogeneity	Robustness to scale variation does not directly resolve the label-dependence structure
Focal/Asymmetric loss variants	[14][13][15]	Imbalance handling and hard-negative reweighting	Improvements at the loss level cannot replace explicit modelling of conditional relations
Imbalanced multi-label evaluation protocols	[16][18]	More reliable comparison under long-tail distributions	Evaluation protocol alone cannot compensate for insufficient model factorization

2.4 Dataset Comparison

MLRSNet is adopted as the primary dataset of this dissertation because it simultaneously provides primary scene categories and secondary multi-label semantics, thereby supporting direct analysis of the conditional decomposition between scene-level and instance-level semantics [1]. This dual-level annotation structure constitutes the methodological basis for selecting MLRSNet as the main experimental benchmark.

Compared with widely used earlier scene datasets, MLRSNet provides stronger support for the study of conditional dependence, long-tail behaviour, and cascade effects. Earlier benchmarks remain valuable for controlled scene-recognition baselines; however, most of them are single-label datasets and are therefore less suitable for validating scene-conditioned multi-label mechanisms [2].

Table 4 compares MLRSNet with representative earlier remote sensing scene datasets.

Table 4. MLRSNet versus earlier remote-sensing scene datasets.

Dataset (MLRSNet first)	Approx. scale	Label structure	Primary use in literature	Relevance to this dissertation
MLRSNet [1]	109,161 images; 46 primary classes; 60 secondary	Joint primary and multi-label secondary annotations	Multi-label semantic scene understanding	Direct correspondence with the two-stage conditional objective

Dataset (MLRSNet first)	Approx. scale	Label structure	Primary use in literature	Relevance to this dissertation
	labels			
UC Merced Land-Use [2]	2,100 images; 21 classes	Single label	Early scene classification benchmark	Provides a useful baseline, but cannot evaluate conditional secondary-label modelling
WHU-RS19 [2]	1,005 images; 19 classes	Single label	Small-sample scene recognition	Limited for the analysis of long-tail effects and multi-label dependence
RSSCN7 [2]	2,800 images; 7 classes	Single label	Scale-variant scene categorization	Useful for scale analysis, but not for hierarchical multi-label learning
AID [2]	10,000 images; 30 classes	Single label	Large-scale scene classification baseline	Supports representation-learning studies, but not scene-conditioned multi-label inference

2.5 Method Comparison

Table 5 compares representative methods from early to recent studies under a unified criterion, namely whether the method jointly addresses dependency modelling, imbalance handling, and alignment with dataset structure.

Table 5. Comparison of representative methods.

Study	Core task formulation	Dependency modeling	Imbalance strategy	Main limitation from this dissertation's perspective
Yang and Newsam [3]	BoVW-based scene classification	None (single-label setup)	Not central	Provides an early reproducible baseline, but does not support multi-label conditional modelling
Hua et al. [6]	Multi-label aerial classification with a relation network	Pairwise label relations	Limited	Models pairwise relations, but does not introduce explicit conditioning on primary scenes
Ji et al. [7]	Latent semantic dependency for multi-label remote sensing	Implicit latent dependencies	Limited	Captures latent dependencies, but the conditional structure remains implicit
Li et al. [8]	CNN + GNN multi-label classification	Graph-based label	Limited	Introduces graph-based interaction, but

Study	Core task formulation	Dependency modeling	Imbalance strategy	Main limitation from this dissertation’s perspective
		interaction		the formulation remains largely unconditional
Wu et al. [9]	Semantic-driven Transformer multi-label classification	Attention-based global dependencies	Indirect	Models global dependencies through attention, but does not explicitly factorize scene hierarchy
Ridnik et al. [14]	Generic multi-label objective design	Not the primary focus	Asymmetric Loss	Addresses gradient imbalance, but not dataset-specific conditional semantics
Park et al. [15]	Long-tailed multi-label objective refinement	Not the primary focus	Robust Asymmetric Loss	Improves robustness under long-tail distributions, but remains an objective-level modification rather than a structural decomposition
This Study	Two-stage $p(y^{pri}, y^{sec} \mid x) = p(y^{pri} \mid x) p(y^{sec} \mid x, y^{pri})$	Explicit scene-conditioned secondary modelling	ASL-GB with step-dependent negative focusing	Designed to align model factorization with the annotation structure of MLRSNet [1][14]

2.6 Gaps and Motivation

The above analysis indicates that the relevant research gaps are not limited to algorithm design, but also concern dataset structure. Existing methods often improve either representation quality or loss robustness, while leaving the conditional factorization insufficiently specified. In addition, existing benchmark practices rely heavily on single-label datasets when evaluating methods intended for structured multi-label semantics.

Table 6 maps each identified gap to a corresponding design choice in this dissertation.

Table 6. Research gaps and dissertation responses.

Gap type	Specific gap in prior work	Evidence in literature	Project-level response in this dissertation
Model gap	Many methods treat multi-label prediction as an unconditional mapping	[6][9][4]	Introduce an explicit two-stage decomposition with primary-scene conditioning
Model gap	Objective improvements and dependency modules are often evaluated separately	[9][14][15]	Use same-backbone controlled comparisons to isolate structural gain from loss gain

Gap type	Specific gap in prior work	Evidence in literature	Project-level response in this dissertation
Model gap	Long-tail reweighting may not adapt across different training stages	[14][15]	Propose ASL-GB with a training-step-dependent negative focusing exponent
Dataset gap	Single-label scene benchmarks cannot validate conditional dependence among secondary labels	[2]	Center the experiments on MLRSNet, which contains joint primary and secondary annotations [1]
Dataset gap	Public dataset descriptions may not fully reflect the complexity of empirical distributions	[1]	Conduct a full dataset audit, including cardinality, co-occurrence, and conditional shift, before model design
Evaluation gap	Aggregate metrics may obscure label-level failure modes	[4][16][18]	Report overall metrics together with label-level analysis and Oracle/Cascade diagnostics

3 Dataset Analysis and Experimental Setup

3.1 Dataset Overview and Statistical Analysis

3.1.1 Dataset Origin and Source

This dissertation uses MLRSNet as the experimental dataset. This dataset is designed for semantic understanding of high-spatial-resolution remote sensing scenes and simultaneously provides primary scene category annotations and secondary multi-label annotations.

Dataset page: https://data.mendeley.com/datasets/7j9bv9vwsx/3

Original paper: Qi Xiaoman et al., *MLRSNet: A multi-label high spatial resolution remote sensing dataset for semantic scene understanding*, ISPRS Journal of Photogrammetry and Remote Sensing, 2020.

There are three reasons for choosing MLRSNet. First, the dataset scale supports stable training and evaluation, and reduces the high variance commonly seen in small-sample remote sensing datasets. Second, the annotation format in which primary categories and secondary labels coexist is consistent with the two-stage decomposition in this dissertation. Third, the data simultaneously contains long-tail imbalance, label co-occurrence, directional dependence, and conditional rearrangement; if it is simplified into a shared backbone followed by 60 mutually independent sigmoid outputs, key conditional structure will be lost.

3.1.2 Dataset Scale and Composition

Table 7 presents the public dataset summary alongside the statistics measured in this dissertation.

Table 7. Public MLRSNet summary and measured statistics.

Item	Public MLRSNet Description	Measured in This Study
Total number of images	109,161	109,161
Number of primary categories	46	46
Number of secondary labels	60	60
Image size	256 X 256	256 X 256
File format	JPG	JPG
Labels per image	Stated as 1-13 in the README	Measured in this study as 0 to 40, with a mean of 5.77

Further statistics show that the mean number of positive secondary labels per image is 5.7707, the minimum is 0, and the maximum is 40; among them, 10 images contain no positive secondary labels, and 3,471 images contain more than 13 positive labels. All images are 256 X 256 pixels, with an average file size of 11.78 KB. These results indicate that the label cardinality distribution of MLRSNet is right-skewed and heavy-tailed, and includes a small number of label-dense samples; therefore, it should not be regarded as a general multi-label task in which “each image contains only a small number of labels.”

The primary label distribution is shown in Figure 1. Among the 46 primary categories, the sample count ranges from 1,500 images (1.37%) for bareland to 2,895 images (2.65%) for basketball_court, with a mean of 2,373.07, a standard deviation of 295.75, and a coefficient of variation of only 0.1246, indicating that the degree of class imbalance in the primary classification task is low. The primary labels are approximately balanced, and therefore Accuracy and Macro-F1 are expected to be numerically close in statistical terms.

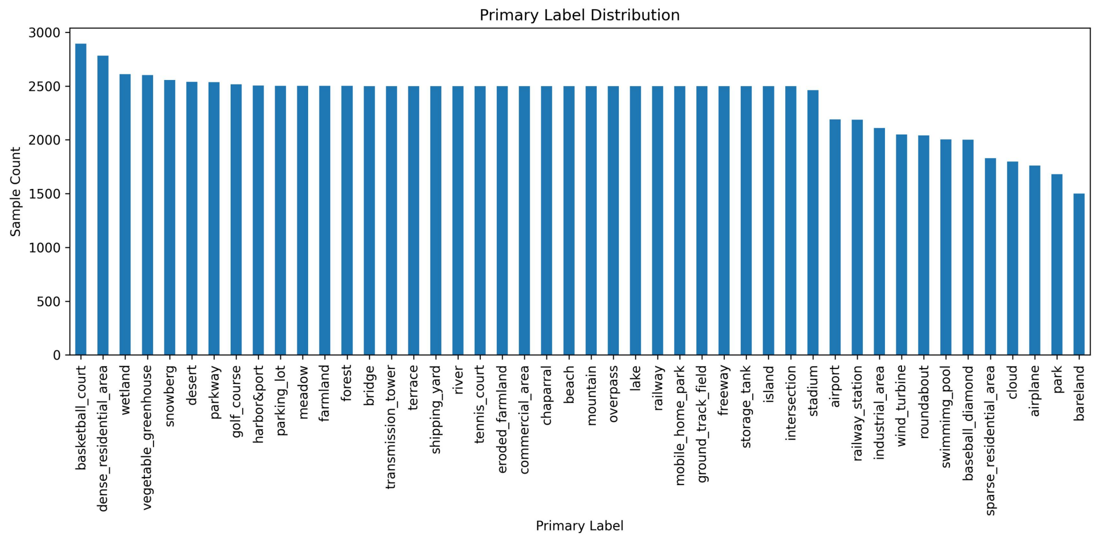

Figure 1. Primary label distribution.

The secondary label distribution is shown in Figure 2 and exhibits a clear long tail. The most frequent labels are trees (64.79%), pavement (51.65%), buildings (47.00%), and grass (45.25%), while the rarest labels are football field (1.02%), cloud (1.69%), baseball diamond (1.83%), and wind turbine (1.88%); among the 60 labels, 33 are below 5%, 49 are below 10%, and only 4 exceed 40%. Therefore, if the secondary task is simplified into the average of 60 independent binary classification tasks with the same level of difficulty, the modelling difficulty of rare labels will be systematically underestimated.

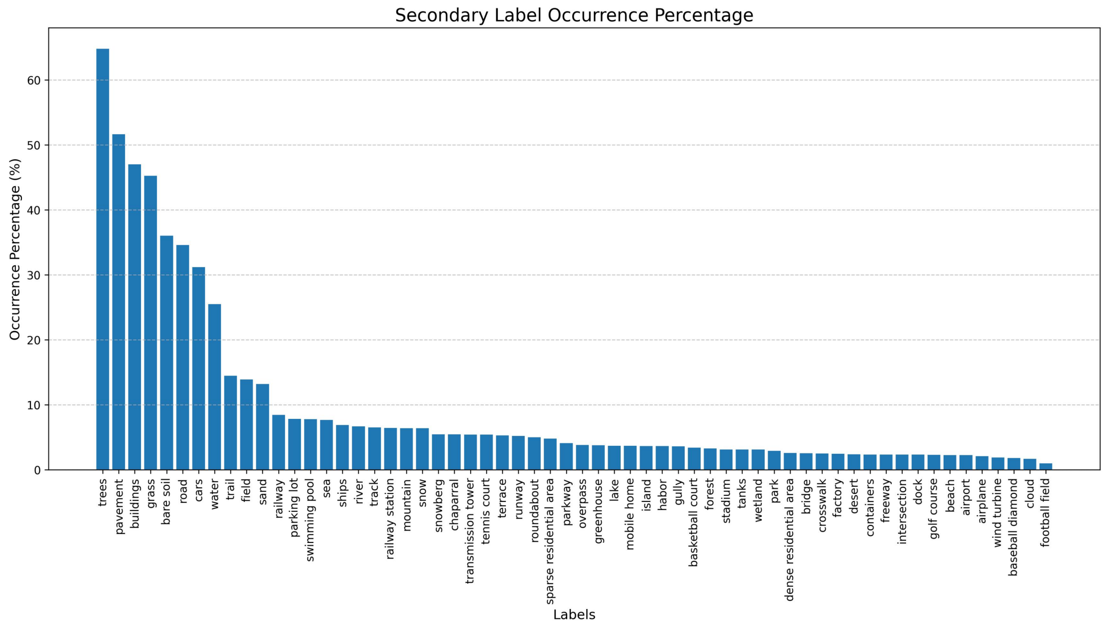

Figure 2. Secondary label frequencies.

3.1.3 Independent-Label Formulation

Let the complete dataset be

$$\mathcal{D} = \{(x_i, p_i, y_i)\}_{i=1}^{N}, \qquad N = 109{,}161, \tag{3.1}$$

where x_i denotes the i-th RGB image, $p_i \in \{0,1\}^{46}$ denotes the one-hot vector of the primary label, satisfying $\| p_i \|_1 = 1$, and $y_i \in \{0,1\}^{60}$ denotes the secondary multi-label vector. The empirical prevalence of secondary label j is defined as

$$\pi_j = \frac{1}{N}\sum_{i=1}^{N} y_{ij}, \tag{3.2}$$

and the empirical probability of primary class k is defined as

$$\rho_k = \frac{1}{N}\sum_{i=1}^{N} p_{ik}. \tag{3.3}$$

For multi-label learning, two basic summary quantities are commonly used and have direct statistical meaning, namely label cardinality and label density **[5]**:

$$\text{Label Cardinality} = \frac{1}{N}\sum_{i=1}^{N} \| y_i \|_1 = 5.7707, \tag{3.4}$$

$$\text{Label Density} = \frac{1}{60N}\sum_{i=1}^{N} \| y_i \|_1 = 0.09618. \tag{3.5}$$

Therefore, among the 60-dimensional secondary label coordinates, about 9.62% are positive and 90.38% are negative on average, so gradients from negative classes dominate during training. Another class-level imbalance metric is the imbalance ratio IRLbl [16]:

$$\text{IRLbl}(j) = \frac{\max_m n_m}{n_j}, \tag{3.6}$$

where $n_j = \sum_i y_{ij}$ denotes the support of label j. Statistics on the current dataset show that the mean imbalance ratio is 16.80, among which football field is the most extreme, with

$$\text{IRLbl(football field)} = \frac{70728}{1108} = 63.83. \tag{3.7}$$

Since the most frequent label trees appears 70,728 times, while football field appears only 1,108 times, the secondary label space has a clear long-tail characteristic. Label cardinality at the sample level is also highly dispersed: its mean is 5.7707, standard deviation is 5.1665, median is 5, and the 95th and 99th percentiles are 10 and 38, respectively, indicating that most images contain a small number of active labels, but there exists a small number of annotation-dense samples.

The balance of primary labels can also be quantified by entropy:

$$H_{\text{primary}} = -\sum_{k=1}^{46} \rho_k \log \rho_k = 3.8206. \tag{3.8}$$

The theoretical maximum for 46 classes is $\log 46 = 3.8286$, and therefore the normalised primary-label entropy is

$$\frac{H_{\text{primary}}}{\log 46} = 0.9979, \tag{3.9}$$

equivalently, the entropy-based effective number of primary classes is

$$\exp(H_{\text{primary}}) = 45.63. \tag{3.10}$$

This is very close to the full number of classes, 46, and provides quantitative support for the approximately balanced distribution of primary labels.

Marginal prevalence alone is still insufficient to explain task difficulty, because the main difficulty of multi-label tasks often comes from the joint distribution. To this end, define the label cardinality of sample i as

$$K_i = \| y_i \|_1 = \sum_{j=1}^{60} y_{ij}. \tag{3.11}$$

If only the marginal prevalence π_j of each label is retained and the 60 secondary labels are assumed to be independent of each other, then K_i follows a Poisson-binomial distribution with parameters $\{\pi_j\}_{j=1}^{60}$, whose mean and variance satisfy

$$\mathbb{E}_{\text{ind}}[K] = \sum_{j=1}^{60} \pi_j = 5.7707, \tag{3.12}$$

$$\text{Var}_{\text{ind}}(K) = \sum_{j=1}^{60} \pi_j (1 - \pi_j) = 4.0897. \tag{3.13}$$

The empirical variance in the data, however, is

$$\widehat{\text{Var}}(K) = 26.6926. \tag{3.14}$$

The ratio of the two reaches

$$\frac{\widehat{\text{Var}}(K)}{\text{Var}_{\text{ind}}(K)} = 6.53, \tag{3.15}$$

which indicates that marginal long-tail behaviour alone is insufficient to explain the fluctuation in label density at the sample level. Further using

$$\text{Var}(K) = \sum_{j=1}^{60} \pi_j (1 - \pi_j) + 2 \sum_{a<b} \text{Cov}(Y_a, Y_b), \tag{3.16}$$

it follows that the 60 labels have overall positive covariance; reversing the above equation gives that the sum of the covariances over all binary label pairs is about 11.30. The multi-label co-occurrence in MLRSNet is therefore a property of the overall distribution rather than a local phenomenon of a small number of samples.

The independence baseline simultaneously overestimates the probability of empty labels and underestimates the proportion of high-cardinality samples. In the real data,

$$\text{P}(K = 0) = 0.0092\%, \qquad \text{P}(K > 13) = 3.1797\%, \tag{3.17}$$

whereas under the independence baseline exactly matched to the marginal prevalence, these probabilities are approximately

$$P_{\text{ind}}(K=0)=0.0875\%,\qquad P_{\text{ind}}(K>13)=0.0313\%. \tag{3.18}$$

Therefore, the right-tail probability of $\mathrm{P}(K>13)$ in the real data is about 102 times that under the independence baseline. Figure 3 shows that the probability mass of the independence baseline is concentrated around 4 to 7 active labels, whereas the empirical distribution is more dispersed in the interval of 2 to 9 labels; in the high-cardinality region, the empirical tail decays more slowly. The joint distribution of MLRSNet is therefore difficult to explain adequately by an independence approximation, and subsequent models need to deal simultaneously with class long tails and label-cluster structure under scene conditions.

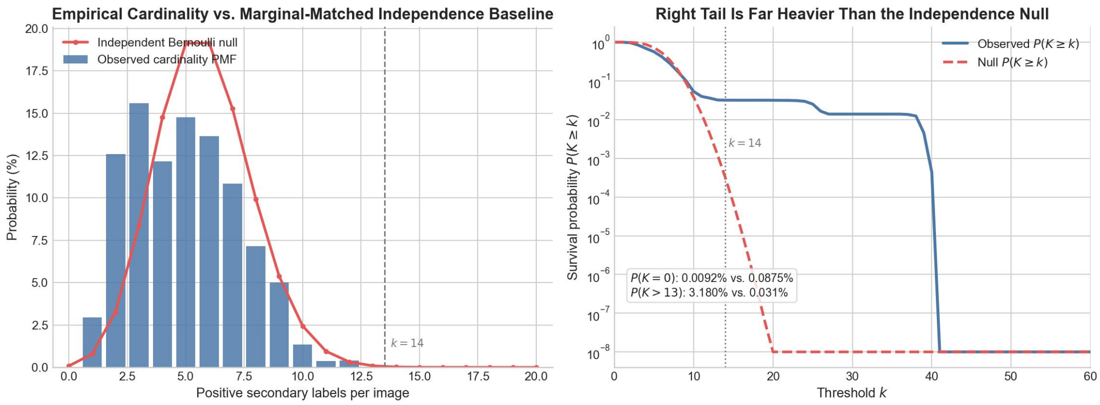

Figure 3. Label-cardinality distribution.

3.2 Stability of Data Splits

This dissertation adopts a split configuration with a fixed random seed of 42, train_ratio = 0.4, and val_ratio = 0.1, with all remaining samples assigned to the test set. The resulting numbers of samples in the training, validation, and test sets are 43,664, 10,916, and 54,581, respectively, corresponding to 40%, 10%, and 50% of the full dataset.

To examine whether random splitting introduces non-negligible structural shift, this dissertation defines split-to-full drift separately for primary and secondary labels:

$$\Delta_{s,k}^{\text{pri}}=\left|\hat{\rho}_{s,k}-\rho_k\right|,\qquad \Delta_{s,j}^{\text{sec}}=\left|\hat{\pi}_{s,j}-\pi_j\right|, \tag{3.19}$$

where $\hat{\rho}_{s,k}$ and $\hat{\pi}_{s,j}$ denote the proportion of primary class k and the prevalence of secondary label j in split s, respectively. Since the current split can be viewed as sampling without replacement from a finite population, if the true proportion of a label in the full dataset is p and the size of the corresponding split is n_s, then the scale of the proportion fluctuation induced by random sampling can be characterised by the hypergeometric approximate standard deviation

$$\sigma(p,n_s)=\sqrt{\frac{p(1-p)(N-n_s)}{(N-1)n_s}} \tag{3.20}$$

Statistics show that across the training, validation, and test sets, the maximum absolute drift and the mean absolute drift of primary label proportions are 0.339 and 0.062 percentage points, respectively; the corresponding values for secondary labels are 0.429 and 0.092 percentage points. Among the primary labels, the maximum drift occurs for freeway in the validation set, with an absolute drift of 0.3389 percentage points, while the corresponding hypergeometric fluctuation scale is about 0.1358 percentage points, or about 2.50 standard deviations; among the secondary labels, the maximum drift occurs for river in the validation set, with an absolute drift of 0.4287 percentage points, while the corresponding scale is about 0.2269 percentage points, or about 1.89 standard deviations.

The results indicate that, although this dissertation does not adopt the iterative stratified splitting method commonly used in multi-label learning [17], the random 40/10/50 split still maintains a high level of consistency in the distributions of primary and secondary labels; the statistical fluctuation of the current split lies within the range of random sampling and is insufficient to become the main source of subsequent experimental results.

3.3 Dependence Between Primary and Secondary Labels

3.3.1 Semantics of the Annotation Scheme

There is a clear but non-one-to-one relationship between the 46 primary categories and the 60 secondary labels. After normalisation, 33 of the 46 primary categories can find directly identical or normalised identical counterparts in the secondary labels, among which the conditional matching rates of golf_course-golf course, wind_turbine-wind turbine, and basketball_court-basketball court are all 1, while baseball_diamond-baseball diamond is 0.997. At the same time, there are also significant heteronymous mappings and misalignments, for example, the conditional matching rates of bareland-bare soil, shipping_yard-containers, and vegetable_greenhouse-greenhouse are 0.995, 1.000, and 1.000, respectively, whereas the conditional matching rate of stadium-stadium is 0.

Primary and secondary labels are neither independent nor simply redundant. According to the conditional probability matrix statistics, the primary label systematically changes the conditional distribution of the secondary labels, while usually not corresponding one-to-one with any single secondary label. Figure 4 gives the conditional probability matrix between primary and secondary labels. If the primary label were merely an alias of a secondary label, then the matrix should mainly take high values near the main diagonal; the actual result is not so. Most cells are close to 0, whereas several rows corresponding to scenes such as intersection, parkway, and dense_residential_area take high values simultaneously in multiple columns. Therefore, in the setting of this dissertation, the primary label should be modelled as a conditioning variable that constrains the entire 60-dimensional secondary distribution, rather than as an indicator variable for a single label.

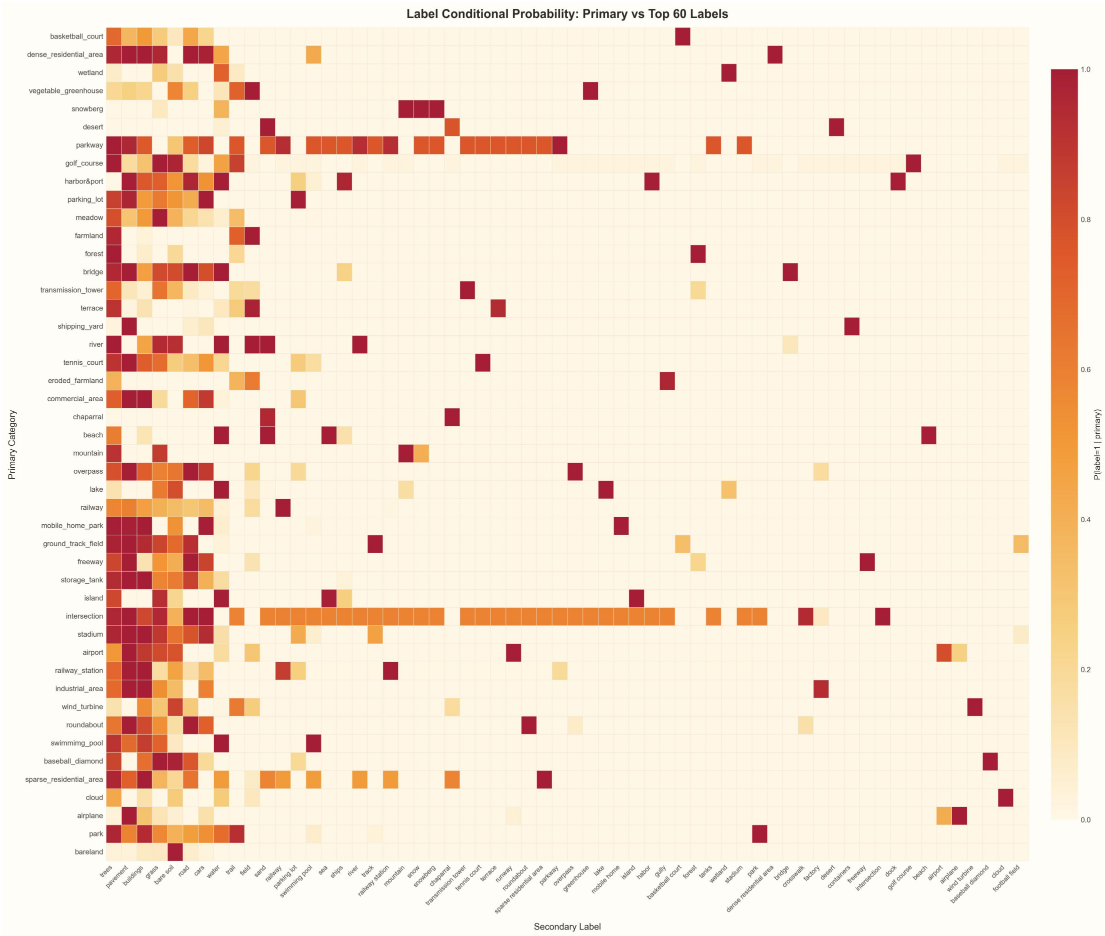

Figure 4. Primary-secondary conditional probabilities.

3.3.2 Label Co-occurrence and Dependency Structure

The secondary label space is not only imbalanced, but also highly structured. For any two secondary labels a and b, let

$$C_{ab} = \sum_{i=1}^{N} y_{ia} y_{ib} \tag{3.21}$$

denote the number of co-occurrences, and let

$$E_{ab}^{(0)} = \frac{n_a n_b}{N} \tag{3.22}$$

denote the expected number of co-occurrences when only marginal prevalence is retained and all remaining structure is treated as independent, where $n_a = \sum_i y_{ia}$ and $n_b = \sum_i y_{ib}$.

Furthermore, define

$$\mathrm{P}(a|\ b) = \frac{C_{ab}}{n_b}, \qquad \mathrm{P}(b|\ a) = \frac{C_{ab}}{n_a}, \tag{3.23}$$

$$\mathrm{Lift}(a,b) = \frac{\mathrm{P}(a,b)}{\mathrm{P}(a)\mathrm{P}(b)}, \tag{3.24}$$

$$J(a,b) = \frac{C_{ab}}{n_a + n_b - C_{ab}}, \tag{3.25}$$

as well as the phi coefficient

$$\phi(a,b) = \frac{P(a,b) - P(a)P(b)}{\sqrt{P(a)(1-P(a))P(b)(1-P(b))}}. \tag{3.26}$$

Among the 60 secondary labels, there are

$$\binom{60}{2} = 1770 \tag{3.27}$$

unordered label pairs in total, of which 518 pairs never co-occur, accounting for 29.27% of all label pairs. This shows that the label co-occurrence graph is sparse, but sparsity does not imply independence. Therefore, observing only raw co-occurrence counts is insufficient to identify dependency structure, because they simultaneously contain information on both prevalence and dependence. Figure 5 gives the conditional probability matrix among secondary labels. Compared with a correlation matrix, this kind of statistic directly preserves directional information: if one column takes high values as a whole under the condition of a given label, while the reverse direction is asymmetric, then conditional subsets, implication relations, or scene dependence can be identified accordingly. The statistics show that high-frequency infrastructure labels such as pavement, road, and cars all have high conditional probabilities under multiple conditions, whereas rare labels such as football field, dock, and runway exhibit local but strong asymmetric conditional dependence. What this statistic reflects is the change in the conditional distribution of the remaining labels given a label, rather than simple label similarity.

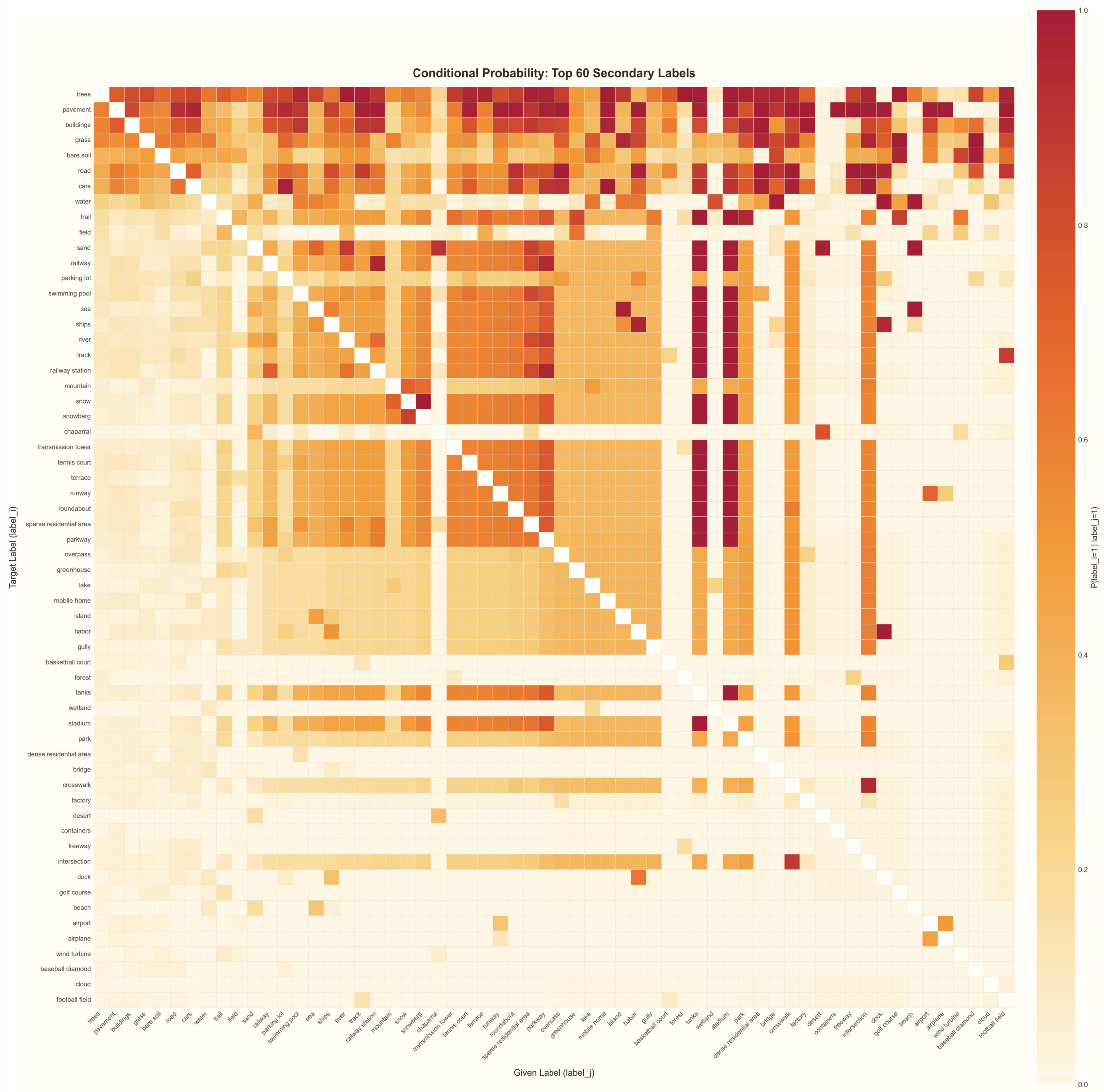

Figure 5. Secondary-label conditional probabilities.

From the perspective of directionality, the dependencies are also clearly asymmetric. For several representative label pairs, dock -> ships satisfies $\mathrm{P(ships|\ dock)} = 0.9449$, but the reverse conditional probability is only 0.3200; parking lot -> cars satisfies $\mathrm{P(cars|\ parking\ lot)} = 0.9796$, while the reverse is only 0.2467; wetland -> water satisfies $\mathrm{P(water|\ wetland)} = 0.7867$, while $\mathrm{P(wetland|\ water)} = 0.0966$. Among all 979 label pairs with co-occurrence counts of no less than 100, 738 pairs have a bidirectional conditional-probability difference of at least 0.1, 510 pairs have a difference of at least 0.2, and 82 pairs reach above 0.8. This shows that many relations in MLRSNet are directed implication relations or conditional subset relations rather than symmetric correlations.

Another phenomenon that needs careful handling is that, in the current label matrix, stadium, tanks, and wetland all have supports of 3,417, and stadium and tanks completely overlap, while neither co-occurs at all with wetland. Such structures are more likely to reflect anomalous patterns caused by annotation conventions. If two labels satisfy $y_{i,a} = y_{i,b}$ for all samples, then any macro-averaged metric of the following form

$$M_{\mathrm{macro}} = \frac{1}{C}\sum_{c=1}^{C} M_c \tag{3.28}$$

will count the same semantic phenomenon twice; therefore, subsequent interpretation of macro-averaged results on the original label set should be combined with this type of structural anomaly.

3.3.3 Information Gain from Conditioning

The previous subsection showed that extensive dependencies exist among secondary labels. Next, the effect of primary labels in reducing uncertainty about secondary labels is further evaluated from an information-theoretic perspective. Let

$$q_{kj} = \mathrm{P}(y_j = 1 | y^{pri} = k), \tag{3.29}$$

and use the binary entropy function

(3.30)

to measure the uncertainty of a single label. Then the amount of information that label j obtains from the primary label is

$$I(Y_j; Y^{pri}) = h(\pi_j) - \sum_{k=1}^{46} \rho_k h(q_{kj}). \tag{3.31}$$

All values below are in bits. Statistics over the 60 secondary labels show that the mean information gain is 0.262 bit and the median is 0.207 bit, indicating that introducing the primary label reduces the uncertainty of the secondary labels.

In terms of absolute information gain, the labels that depend most strongly on the primary label include pavement (0.817 bit), road (0.604 bit), cars (0.577 bit), buildings (0.557 bit), and water (0.556 bit). These labels themselves occur frequently, but their occurrence still clearly depends on the scene, and they can therefore be grouped as high-frequency labels significantly affected by scene conditions. On the other hand, in terms of the relative entropy reduction ratio, runway, dock, factory, and wind turbine eliminate 83.9%, 97.3%, 78.7%, and 99.9% of their label uncertainty from the primary label, respectively, indicating that although these labels are low in absolute frequency, their occurrence probability is largely determined by scene conditions.

Discussing only single-label information gain is still insufficient, because the primary label also changes the shape of the whole 60-dimensional secondary label vector. To this end, for each primary scene k, define the overall rearrangement strength under the scene condition as

$$D_k = \sum_{j=1}^{60} \mathrm{KL}\Big(\mathrm{Bern}(q_{kj}) \| \mathrm{Bern}(\pi_j)\Big). \tag{3.32}$$

This quantity measures the distance between the 60-dimensional secondary marginal distribution conditioned on a given primary label and the global unconditional distribution. Statistics show that the mean of D_k is 15.57 bits and the median is 13.42 bits, among which the scenes intersection and parkway have the highest values, reaching 63.54 bits and 56.91 bits, respectively. For these primary classes, directly using the global unconditional distribution for second-stage inference would produce a clear prior mismatch.

Likewise, the primary scene also significantly changes the expected number of active labels. For

$$\mathbb{E}[\| y^{sec} \|_1 | y^{pri} = k] = \sum_{j=1}^{60} q_{kj} \tag{3.33}$$

statistics show that its value ranges from 1.34 for bareland to 25.80 for intersection, a span of nearly 19.3 times; the global mean is 5.77, while the median under primary-scene conditioning is 5.16, with an interquartile interval of [3.40, 6.55]. The primary label therefore significantly changes the conditional label cardinality. Figure 6 shows, on the left, the expected label cardinality under different primary labels, and, on the right, the secondary labels with the highest information gain from the primary label. The results show that, besides low-frequency labels such as runway, dock, and wind turbine, high-frequency labels such as pavement, road, cars, and buildings are also sensitive to primary-scene conditions. Therefore, the information gain of the primary label is reflected not only in low-frequency labels, but also in common labels.

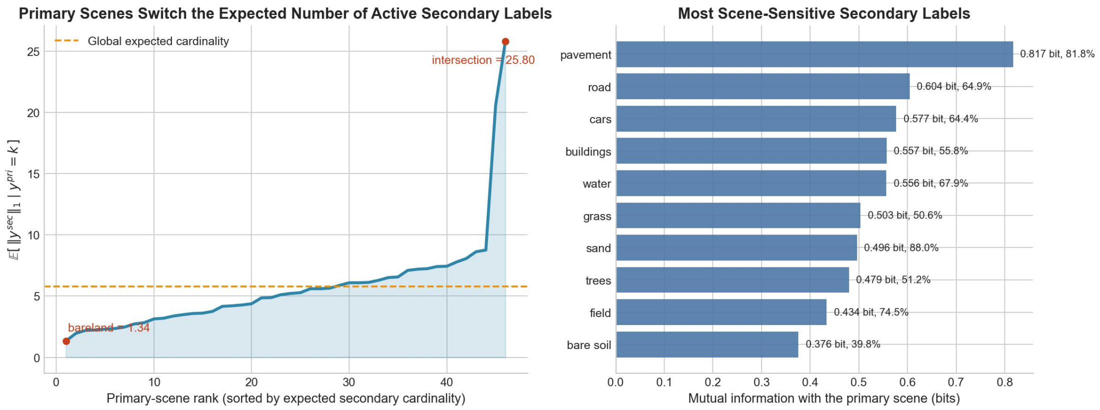

Figure 6. Conditioning information gain.

3.3.4 Higher-Order Dependency Structure

Table 8 gives representative rearrangements of secondary-label prevalence under primary-label conditions.

Table 8. Secondary-label rearrangements under primary-scene conditions.

primary scene	secondary label	$P(\text{label} \mid \text{primary})$	global $P(\text{label})$	Lift
airport	runway	99.54%	5.20%	19.14
harbor&port	dock	99.48%	2.33%	42.70
industrial_area	factory	93.74%	2.49%	37.65
vegetable_greenhouse	field	100.00%	13.92%	7.18
wind_turbine	wind turbine	99.95%	1.88%	53.25
dense_residential_area	cars	98.78%	31.21%	3.17

Statistics over all $46 \times 60 = 2760$ primary-label-secondary-label combinations show that 2,298 combinations satisfy $P(\text{label} \mid \text{primary}) \le 1\%$, 272 combinations satisfy $P(\text{label} \mid \text{primary}) \ge 50\%$, and the numbers of combinations satisfying $\text{lift} \ge 2/5/10/20/40$ are 209, 127, 88, 33, and 11, respectively. These statistics show that the primary label significantly changes the conditional support region of secondary labels, which is why Chapter 4 adopts a conditional decomposition.

From the perspective of higher-order combinations, conditional rearrangement still exists under higher-order conditional combinations. Taking cars as an example:

$$\mathrm{P(cars|\ road)} = 65.37\%, \quad \mathrm{P(cars|\ road, pavement)} = 69.07\%,$$

$$\mathrm{P(cars)} = 31.21\%, \mathrm{P(cars|\ buildings, road, pavement)} = 70.65\%. \tag{3.34}$$

This shows that buildings -> road -> pavement -> cars forms a typical multi-hop conditional dependency structure. On the other hand, higher-order correlation does not imply that the more conditions are added, the larger the conditional probability becomes. For example,

$$\mathrm{P(airplane|\ airport)} = 46.67\%, \quad \mathrm{P(airplane|\ airport, runway)} = 24.27\%. \tag{3.35}$$

Such counterexamples indicate that condition-adaptive label-correlation modelling cannot simply impose a uniform positive gain on correlation terms only on the basis of label co-occurrence; instead, the probability distribution needs to be re-estimated under different conditional combinations. Therefore, what needs to be learned in MLRSNet is not a static co-occurrence table, but a conditional distribution jointly determined by the primary scene, local evidence, and label interactions.

3.3.5 Implications for Model Design

Based on the above analysis, four important statistical implications can be obtained:

1. The primary-label distribution is approximately balanced, and therefore the Accuracy, Micro-F1, and Macro-F1 of the first stage should be statistically close; if the three are clearly differentiated in Chapter 6, this indicates that the errors are concentrated in a small number of specific primary classes; otherwise, their closeness is consistent with the conclusion of this chapter.
2. The primary label provides stable information gain for the secondary labels, and the scene-conditioned distribution exhibits a significant KL deviation from the global distribution. Therefore, adopting the decomposition $p(y^{pri}, y^{sec}|\ x) = p(y^{pri}|\ x)p(y^{sec}|\ x, y^{pri})$ is more consistent with the statistical structure in the data than modelling all 60 secondary labels as unconditional independent outputs.
3. The variance of label cardinality is inflated by 6.53 times relative to the independence baseline, and the right-tail probability increases by about 102 times, indicating that the difficulty of the secondary task lies not only in the marginal long tail, but also in scene heterogeneity and positively correlated cluster structure in the joint distribution.
4. Labels with high lift and high mutual information are usually sensitive to primary-scene conditions, and therefore misclassification of the primary label should not be regarded as independent additional noise, but should be incorporated into the analysis of structured cascade degradation.

3.4 Experimental Dataset Setup

3.4.1 Experimental Data Preparation

Data preparation includes sample-index construction, primary-label encoding, dataset splitting, and image tensorisation and normalisation, without involving additional image restoration or sample filtering. Data reading generates image paths, primary labels, and secondary labels from the 46 primary category directories and their annotation files, and constructs a 46-dimensional one-hot representation of the primary label for each sample, so that it can be used both for first-stage single-label classification and as the conditional input to the second stage; the data are then split using the fixed random seed 42, and input transformation includes only RGB conversion, tensorisation, and normalisation.

Image processing does not additionally introduce augmentation strategies such as resizing, cropping, random flipping, colour jitter, or Mixup, but applies only numerical normalisation while preserving the original spatial layout. This choice is based on the following considerations: the main statistical difficulty of this task lies not in differences in input scale, but in label rearrangement under scene conditions, long-tail negative classes, and structured dependence; excessive geometric operations or resampling may change the spatial layout on which remote sensing scenes highly depend; at the same time, adopting a minimal preprocessing setting is also conducive to attributing performance differences mainly to model structure and loss design rather than to additional data augmentation strategies. Moreover, this is also the setting recommended by the official DINOv3 repository.

3.4.2 Compatibility with Preprocessing

First, this dissertation does not perform manual label cleaning or label merging. The reason is not that all annotations are assumed to be error-free by default, but that Section 3.3 has already shown that the label matrix simultaneously contains heteronymous mappings, conditional subsets, near-deterministic hierarchical dependencies, and anomalous overlaps. On the basis of name similarity, close support counts, or subjective semantic judgement alone, it is difficult to reliably decide, in the absence of additional evidence, which labels should be deleted, merged, or renamed; directly performing these operations may mistakenly remove information belonging to the data structure as if it were noise.

Second, this dissertation does not adopt rare-label deletion or simple resampling. Sections 3.1 and 3.3 have already shown that rare labels cannot uniformly be regarded as low-frequency noise. Some of them simultaneously have high lift and high relative information gain, such as wind turbine, dock, runway, and intersection. The conditional statistics corresponding to these labels are an important source of information from the primary label. If labels are deleted or oversampled only because their support is low, important conditional relationships such as football field -> pavement, snowberg -> snow, and harbor&port -> dock may be distorted.

Finally, during training and standard evaluation, the secondary task uses the true primary label as the condition, rather than directly using the prediction of the primary classifier. In this way, the performance level that

$$p(y^{sec} \mid x, y^{pri}) \tag{3.36}$$

can achieve under the true primary condition can first be estimated; otherwise, errors in the primary task and errors in the secondary task would be mixed together, making it difficult to distinguish the source of gains.

3.4.3 Final Experimental Configuration

Table 9 summarises the final dataset configuration used in this dissertation.

Table 9. Final dataset configuration.

Component	Final value
Total number of images	109,161
Number of primary categories	46
Number of secondary labels	60
Train / validation / test split	43,664 / 10,916 / 54,581

Component	Final value
Random seed	42
Number of deleted samples	0
Number of deleted or merged labels	0
Input size	256 X 256 RGB
Output dimension of primary task	46
Output dimension of secondary task	60
Conditioning vector of the multi-label model	46-dimensional primary-label one-hot
Condition for standard secondary evaluation	Ground-truth primary
Condition for cascade error analysis	Predicted primary (Chapter 6 only)

4 Proposed Model Design

4.1 Problem Definition

According to the statistical results in Chapter 3, among all 46 X 60 = 2,760 primary-secondary combinations, 2,298 combinations satisfy $\mathrm{P(label|\ primary)} \le 1\%$, while another 272 combinations satisfy $\mathrm{P(label|\ primary)} \ge 50\%$.

Based on these statistical characteristics, this dissertation adopts the following probabilistic decomposition:

$$p(y^{pri}, y^{sec} \mid x) = p(y^{pri} \mid x) p(y^{sec} \mid x, y^{pri}). \tag{4.1}$$

Accordingly, the task is defined in this dissertation as a two-stage task consisting of scene discrimination and conditional multi-label inference, in which the primary label serves as the conditioning variable of the secondary distribution. In implementation, the two stages each use independently trained DINOv3 models: the first stage estimates $p(y^{pri} \mid x)$, and the second stage estimates $p(y^{sec} \mid x, y^{pri})$ by combining patch tokens and the primary one-hot condition in a conditional label-correlation head.

4.2 Proposed Model Architecture

4.2.1 Primary Classification Module

The first stage uses a DINOv3 ViT-S/16 distilled backbone as the primary scene classifier. This backbone is initialised with public pre-trained weights, and a 46-dimensional linear classifier is attached after its global CLS representation. Let the DINOv3 encoder be denoted by f_{DINO} , then the visual representation in the first stage satisfies

$$v(x) = f_{\mathrm{DINO}}(x) \in \mathbb{R}^{B \times 384}, \tag{4.2}$$

and the corresponding primary logits are

$$z^{pri} = v(x) W_{pri} + b_{pri} \in \mathbb{R}^{B \times 46}. \tag{4.3}$$

There are two main reasons for this design. First, remote sensing scene recognition depends on global spatial layout and long-range contextual structure, and both types of information can be represented by the global representation of a vision Transformer [10][12]. Second, keeping the structure of the first stage simple facilitates subsequent cascade error analysis, because the output of the first stage is an explicit scene prior rather than an additionally designed complex scene-specific head.

4.2.2 Conditional Correlation Module

As shown in Figure 7, the second stage is not implemented as 60 mutually independent sigmoid classifiers on a single global DINOv3 feature. Instead, it follows a structured conditional pipeline with three coupled parts. First, dense patch tokens are projected to T_x , the primary one-hot condition c is linearly projected and reshaped into $R = 8$ condition tokens T_c , and the joint context is formed by concatenation as $X = [T_x, T_c]$. Second, 60 shared learnable label prototypes are expanded into the initial label state and processed by $L = 3$ stacked label-correlation blocks, each containing label self-attention, label-to-context cross-attention, and a feed-forward update, with residual-plus-normalisation updates after each sub-step, to produce the conditional label state $E^{(L)}$. Third, the final label state $E^{(L)}$ queries projected image tokens to obtain label-wise spatial evidence S; then $\left[E_i^{(L)} \| S_i\right]$ is fused by a shared readout function and combined with a per-label bias to produce the 60 secondary logits. The basis of this structure is that secondary label correlations need to be modelled under the joint constraint of image content and the primary condition [6][9][4].

The concrete configuration used in the experiments of this dissertation is as follows:

Table 10 gives the concrete instantiated configuration of the second-stage model.

Table 10. Instantiated second-stage model configuration.

Component	Value
Number of secondary labels N	60
Condition dimension D_c	46
Backbone output dimension F	384
Label embedding dimension d	384
Number of patch tokens M	256
Number of condition tokens R	8
Number of attention heads H	8
Number of reasoning layers L	3
Length of cross context	264
Head dropout	0.1

The total number of parameters of the secondary model is 29,903,293, among which the DINOv3 backbone accounts for 21,601,152 parameters and the conditional label-correlation head accounts for 8,302,141 parameters. This indicates that the added capacity is mainly concentrated in the conditional relation-reasoning and readout components, which transform dense visual features into condition-aware label interaction and label-specific evidence extraction. From the perspective of the overall functional form, the logit of the i-th label is jointly determined by the conditional label state and spatial evidence:

$$z_i(x,c) = g\left([E_i^{(L)}(x,c) \| S_i(x,c)]\right) + b_i, \tag{4.4}$$

where $E_i^{(L)}(x,c)$ is the final representation of the i-th label after three layers of conditional relation reasoning, $S_i(x,c)$ is the spatial evidence extracted for that label from the patch-token sequence, and $g(\cdot)$ denotes the shared MLP. In contrast to the independent sigmoid head

$$z_i^{\text{ind}} = w_i^{\top} v(x) + b_i, \tag{4.5}$$

the discriminative function here varies jointly with image content, label interaction state, and the primary condition. This is the main difference between the structured conditional design in Figure 7 and an independent output head.

4.2.3 Formal Model Definition

This section gives the tensor definitions for the context-construction part that supports downstream relation reasoning and readout in Figure 7. The second stage takes the image x and the primary condition c as input, where $c \in \{0,1\}^{B\times 46}$ is the one-hot scene vector. When training the second stage, c takes the ground-truth primary label; in the cascade inference setting, c is converted from the prediction of the first stage into a one-hot condition. The DINOv3 backbone first outputs a dense patch-token sequence; for an input of 256 × 256 and a patch size of 16 × 16, one obtains $M = 256$ patch tokens. Let the backbone output be

$$\tilde{T}_x = f_{\text{patch}}(x) \in \mathbb{R}^{B\times M\times F}, \tag{4.6}$$

where $F = 384$. The patch-token projection in the head is written as

$$T_x = \tilde{T}_x W_x + b_x \in \mathbb{R}^{B\times M\times d}, \tag{4.7}$$

where under the setting of this dissertation $d = 384$. The implemented model also supports optional token subsampling, but the experimental configuration used here retains all 256 projected patch tokens. The primary condition is then passed through the conditional projection and reshaped to generate the condition token:

$$T_c = \text{reshape}(cW_c + b_c) \in \mathbb{R}^{B\times R\times d}, \tag{4.8}$$

where $R = 8$. For the 60 secondary labels, the model maintains a globally shared learnable label-prototype matrix:

$$E_{\text{base}} \in \mathbb{R}^{N\times d}, \qquad N = 60. \tag{4.9}$$

For the b-th sample in the batch, its initial label state satisfies

$$E_b^{(0)} = E_{\text{base}}, \qquad b = 1,\ldots,B, \tag{4.10}$$

therefore $E^{(0)} \in \mathbb{R}^{B\times N\times d}$. The three main types of representations in the second stage are the patch token T_x, the condition token T_c, and the label token $E^{(0)}$. Among them, the patch token and the condition token jointly form the joint context required for relation reasoning:

$$X = [T_x, T_c] \in \mathbb{R}^{B\times (M+R)\times d}. \tag{4.11}$$

Under the experimental configuration of this dissertation,

$$\begin{gathered} E_{\text{base}} \in \mathbb{R}^{60\times 384}, \quad E^{(0)} \in \mathbb{R}^{B\times 60\times 384}, \quad \tilde{T}_x \in \mathbb{R}^{B\times 256\times 384}, \\ T_x \in \mathbb{R}^{B\times 256\times 384}, \quad T_c \in \mathbb{R}^{B\times 8\times 384}, X \in \mathbb{R}^{B\times 264\times 384} \end{gathered} \tag{4.12}$$

The correspondence between these tensors and Figure 7 is direct: the DINOv3 backbone outputs $\tilde{T}_x$, the patch projection produces T_x, the conditional projection maps c to T_c, concatenation forms $X = [T_x, T_c]$, and $E^{(0)}$ is the initial label state entering the stacked relation blocks.

Figure 7 presents a schematic overview of the overall structure of the second stage.

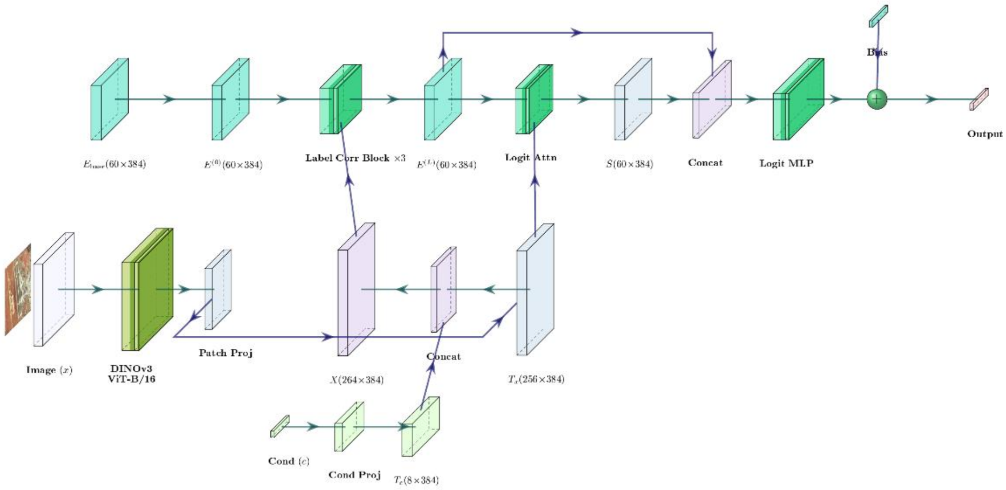

Figure 7. Second-stage architecture.

4.2.4 Computational Pipeline

Combined with Figure 7, the computation of the second stage can be written in a direct model form. Section 4.2.3 has already defined context construction: the patch token T_x is obtained from the image, the primary one-hot vector is projected to the condition token T_c, and the two are concatenated into the joint context $X=[T_x,T_c]$. The relation module is composed of $L=3$ blocks connected in series. Let the input to layer l be $E^{(l-1)}$. The self-attention sub-step is

$$E_{\text{mid}}^{(l)}=\mathrm{LN}\Big(E^{(l-1)}+\mathrm{MHA}(E^{(l-1)},E^{(l-1)},E^{(l-1)})\Big), \tag{4.13}$$

$$E_{\text{ctx}}^{(l)}=\mathrm{LN}\Big(E_{\text{mid}}^{(l)}+\mathrm{MHA}(E_{\text{mid}}^{(l)},X,X)\Big), \tag{4.14}$$

$$E^{(l)}=\mathrm{LN}\Big(E_{\text{ctx}}^{(l)}+\mathrm{FFN}(E_{\text{ctx}}^{(l)})\Big). \tag{4.15}$$

This corresponds to the update order shown in Figure 8: self-attention, residual-normalisation update, cross-attention, residual-normalisation update, feed-forward update, and residual-normalisation update. In the full model, the same block is stacked three times by the relation stack shown in Figure 7.

In the final readout part, spatial evidence is extracted by using the final label state as the query and the projected image tokens as the key/value:

$$S=\mathrm{MHA}(E^{(L)},T_x,T_x)\in\mathbb{R}^{B\times N\times d}, \tag{4.16}$$

then the final label state and spatial evidence are concatenated, the shared readout head generates one scalar for each label, and the learnable bias is added:

$$z_i=g\Big([E_i^{(L)}\|S_i]\Big)+b_i,\qquad i=1,\ldots,N. \tag{4.17}$$

The model head therefore outputs logits rather than probabilities. When probabilities are needed for loss computation or evaluation, sigmoid is applied outside the model:

$$\hat{p}_i=\sigma(z_i),\qquad i=1,\ldots,N. \tag{4.18}$$

Therefore, Figure 7 can be summarised as follows: context construction first maps c to T_c and concatenates it with T_x to form X; stacked relation reasoning produces the conditional label state $E^{(L)}$; readout combines $E^{(L)}$ with image-token evidence and outputs the secondary logits.

4.2.5 Label Correlation Structure

Figure 8 visualises the internal structure of one label-correlation block; the same block is stacked three times in the actual model, exactly as indicated by the relation stack in Figure 7. The first sub-operation is label self-attention. In this step, label correlation is not a pre-given co-occurrence matrix, but is dynamically generated by self-attention on the current sample. For the h-th attention head of layer l,

$$Q_h^{(l)} = E^{(l-1)} W_{Q,h}, \qquad K_h^{(l)} = E^{(l-1)} W_{K,h}, \qquad V_h^{(l)} = E^{(l-1)} W_{V,h},$$

$$R_h^{(l)} = \mathrm{softmax}_{\mathrm{row}}\left(\frac{Q_h^{(l)}(K_h^{(l)})^\top}{\sqrt{d_h}}\right), \qquad O_h^{\mathrm{self},(l)} = R_h^{(l)} V_h^{(l)}. \tag{4.19}$$

Here $R_h^{(l)} \in \mathbb{R}^{B\times N\times N}$, and for each label-query position, $\sum_{j=1}^{N} R_{h,ij}^{(l)} = 1$. This attention weight matrix is the basis of the self-attention sub-operation in Figure 8. Since it is computed from the label state of the current sample rather than fixed in advance by dataset statistics, the model learns sample-adaptive label relations. Moreover, since attention generally does not satisfy

$$R_{h,ij}^{(l)} \neq R_{h,ji}^{(l)}, \tag{4.20}$$

this relation is generally directed and asymmetric.

The second sub-operation uses $E_{\mathrm{mid}}^{(l)}$ as the query and the joint context $X = [T_x, T_c]$ as the key/value, corresponding to the cross-attention sub-operation in Figure 8:

$$Q_h^{\mathrm{cross},(l)} = E_{\mathrm{mid}}^{(l)} W_{Q,h}^{\mathrm{cross}}, \qquad K_h^{\mathrm{cross},(l)} = X W_{K,h}^{\mathrm{cross}}, \qquad V_h^{\mathrm{cross},(l)} = X W_{V,h}^{\mathrm{cross}},$$

$$A_h^{\mathrm{cross},(l)} = \mathrm{softmax}_{\mathrm{row}}\left(\frac{Q_h^{\mathrm{cross},(l)}(K_h^{\mathrm{cross},(l)})^\top}{\sqrt{d_h}}\right), \qquad O_h^{\mathrm{cross},(l)} = A_h^{\mathrm{cross},(l)} V_h^{\mathrm{cross},(l)}. \tag{4.21}$$

Here $A_h^{\mathrm{cross},(l)} \in \mathbb{R}^{B\times N\times (M+R)}$, and for each label-query position, $\sum_{u=1}^{M+R} A_{h,iu}^{\mathrm{cross},(l)} = 1$. Since $X = [T_x, T_c]$ is formed by concatenating the patch token and the condition token, the cross matrix can further be written in blocks according to source:

$$A_h^{\mathrm{cross},(l)} = \left[A_{h,x}^{(l)} \mid A_{h,c}^{(l)}\right], \qquad A_{h,x}^{(l)} \in \mathbb{R}^{B\times N\times M}, \quad A_{h,c}^{(l)} \in \mathbb{R}^{B\times N\times R}. \tag{4.22}$$

Therefore, the weight allocation of cross-attention simultaneously covers image evidence and the primary-scene condition.

The third sub-operation inside the same block is the feed-forward update after cross-attention:

$$E^{(l)} = \mathrm{LN}\big(E_{\mathrm{ctx}}^{(l)} + \mathrm{FFN}(E_{\mathrm{ctx}}^{(l)})\big). \tag{4.23}$$

Hence Figure 8 should be read as one relation block with three internal sub-operations: self-attention, cross-attention, and feed-forward update, each followed by the corresponding residual-normalisation update.

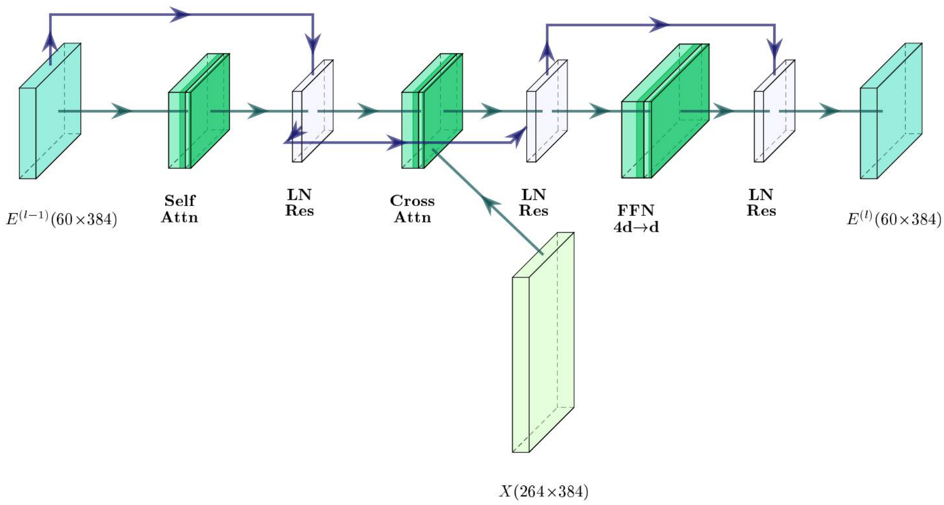

Figure 8. Label-correlation block.

4.2.6 Multi-Layer Correlation Modelling and Output Generation

The actual model composes three such label-correlation blocks, as shown by the relation stack in Figure 7. If the l-th relation block is denoted by Φ_l, then the final label state after stacking three layers can be written as

$$E^{(L)} = \Phi_L \circ \cdots \circ \Phi_1(E^{(0)}; T_x, T_c). \tag{4.24}$$

Since each Φ_l successively includes label-relation propagation, conditional-context injection, and the FFN update, $E^{(L)}$ contains the label state after the joint action of image content and the primary condition. Therefore, the three-layer relation stack in Figure 7 represents increased relation-propagation depth; when pointwise nonlinearity is ignored, multi-layer stacking can represent higher-order dependencies through composite propagation terms similar to $R^{(2)}R^{(1)}$.

The final prediction head is still label-wise at the logit level:

$$z_i = g\left([E_i^{(L)} \| S_i]\right) + b_i, \qquad i = 1, \ldots, N. \tag{4.25}$$

When probabilities are required, sigmoid is applied outside the model:

$$\hat{p}_i = \sigma(z_i), \qquad i = 1, \ldots, N. \tag{4.26}$$

Then the conditional multi-label probability in the second stage can be written as

$$p_\theta(y^{sec} \mid x, c) \approx \prod_{i=1}^{N} \hat{p}_i^{y_i} \left(1 - \hat{p}_i\right)^{1-y_i}. \tag{4.27}$$

Here the model output is still converted into label-wise probabilities by sigmoid, but the correlation is not explicitly parameterised in a structured output layer; instead, it is encoded in the shared conditional label state $E^{(L)}$ and the label-specific spatial evidence S.

5 ASL-GB Loss Design and Analysis

5.1 Loss Function Definition

The first-stage primary scene classification model uses BCE loss. In the second stage, focal-type hard-example reweighting follows Focal Loss [13], while asymmetric focusing and negative-side clipping follow the ASL line [14], [15]. Related long-tail objective designs, including class-balanced reweighting, margin-aware balancing, and logit adjustment, provide further background for imbalance handling [19][20]. This dissertation further generalises the fixed negative-side focusing exponent into a training-step-dependent parameter $\gamma_{-,t} \in [\gamma_{\min}, \gamma_{\max}]$, denoted as ASL-GB. The loss definition and notation are first given below, followed by an explanation of the update criterion for $\gamma_{-,t}$.

Let the mini-batch corresponding to index t be

$$\mathcal{B}_t = \{(x_b, y_b^{pri}, y_b^{sec})\}_{b=1}^{B}, \tag{5.1}$$

where $y_b^{pri} \in \{1, \ldots, K_{\text{pri}}\}$ is the index of the primary class, $y_b^{sec} \in \{0,1\}^{K_{\text{sec}}}$ is the secondary multi-label vector, and let

$$y_{bi} := (y_b^{sec})_i, \qquad i = 1, \ldots, K_{\text{sec}}. \tag{5.2}$$

Let e_k be the standard basis vector, then the conditioning vector used in the secondary stage is

$$c_b := e_{y_b^{pri}} \in \{0,1\}^{K_{\text{pri}}}. \tag{5.3}$$

For the secondary conditional multi-label model, let its logit mapping be

$$f_\phi : \mathcal{X} \times \{0,1\}^{K_{\text{pri}}} \to \mathbb{R}^{K_{\text{sec}}}, \qquad z_b = f_\phi(x_b, c_b), \qquad p_{bi} = \sigma(z_{bi}), \quad i = 1, \ldots, K_{\text{sec}}. \tag{5.4}$$

Fix the positive-side focusing exponent $\gamma_+ > 0$, the clip constant $c \in (0,1)$, and the feasible interval of the negative-side exponent

$$0 < \gamma_{\min} \le \gamma_{\max}, \tag{5.5}$$

and define the negative-side margin variable

$$m_{bi} := (p_{bi} - c)_+ = (\max\{p_{bi} - c, 0\}). \tag{5.6}$$

Then, for any fixed negative-side focusing exponent $\gamma \in [\gamma_{\min}, \gamma_{\max}]$, the empirical risk of the secondary inner layer on the current batch is defined as [14][13][15]

$$\mathcal{J}_t(\phi; \gamma) = \frac{1}{BK_{\text{sec}}} \sum_{b=1}^{B} \sum_{i=1}^{K_{\text{sec}}} \Big[-y_{bi}(1-p_{bi})^{\gamma_+} \log p_{bi} - (1-y_{bi}) m_{bi}^{\gamma} \log(1-m_{bi}) \Big]. \tag{5.7}$$

Here normalisation is carried out by arithmetic averaging over all coordinates of the whole secondary tensor.

If the negative-side focusing exponent takes the value $\gamma_{-,t} \in [\gamma_{\min}, \gamma_{\max}]$, then the corresponding secondary objective function is

$$\mathcal{L}_{\sec} = \mathcal{J}_t(\phi;\gamma_{-,t}) = \frac{1}{BK_{\sec}} \sum_{b=1}^{B}\sum_{i=1}^{K_{\sec}} \left[- y_{bi}(1-p_{bi})^{\gamma_+} \log p_{bi} - (1-y_{bi})(1-\tilde{p}_{bi})^{\gamma_{-,t}} \log \tilde{p}_{bi} \right]$$

(5.8)

where

$$\tilde{p}_{bi} := 1 - m_{bi} = 1 - (p_{bi} - c)_+. \tag{5.9}$$

Correspondingly, the coordinate-wise negative branch can be written as

$$\ell_\gamma^-(p) := -(p-c)_+^\gamma \log\big(1-(p-c)_+\big), \qquad p \in (0,1). \tag{5.10}$$

Since $c \in (0,1)$ and $\gamma > 0$, when $p_{bi} \le c$ the negative branch is exactly zero; when $p_{bi} > c$, the negative branch can be written as a power-weighted logarithmic term on the interval $(c,1)$. During training, $\gamma_{-,t}$ is not pre-given externally, but is adaptively determined by the matching objective of positive and negative statistics on the current batch; within a given training step, it can be regarded as an exogenous scalar with respect to the model parameter ϕ.

5.1.1 Update Rule for $\gamma_{-,t}$

Unlike the original ASL, which updates based on the heuristic probability gap Δp, this dissertation determines $\gamma_{-,t}$ by batch-level matching of positive and negative statistics. Compared with methods that mainly rely on dataset-level class priors or static frequency corrections [19][20], this mechanism updates the negative-side exponent online from current-batch statistics. Let the numbers of positive and negative coordinates in the current batch be

$$N_t^+ := \sum_{b=1}^{B}\sum_{i=1}^{K_{\sec}} y_{bi}, \qquad N_t^- := \sum_{b=1}^{B}\sum_{i=1}^{K_{\sec}} (1-y_{bi}), \tag{5.11}$$

and take the numerical stability constant $0 < \varepsilon < 1$. Define the batch statistic of the positive branch as

$$G_t^+ := \frac{1}{N_t^+ + \varepsilon} \sum_{b=1}^{B}\sum_{i=1}^{K_{\sec}} y_{bi}(1-p_{bi})^{\gamma_+}(1-p_{bi}). \tag{5.12}$$

It retains the focal weight $(1-p_{bi})^{\gamma_+}$ and additionally contains the factor $(1-p_{bi})$, used to characterise the average update strength of the positive branch. For the negative branch, let

$$a_{bi} := \mathbf{1}\{m_{bi} > 0\}, \qquad m_{bi,\varepsilon} := \max\{m_{bi}, \varepsilon\}, \tag{5.13}$$

then the corresponding batch statistic of the negative branch is

$$G_t^-(\gamma) := \frac{1}{N_t^- + \varepsilon} \sum_{b=1}^{B}\sum_{i=1}^{K_{\sec}} (1-y_{bi}) \left[\gamma m_{bi,\varepsilon}^{\gamma-1}\big(-\log(1-m_{bi})\big) + \frac{m_{bi}^\gamma}{1-m_{bi}+\varepsilon} \right] p_{bi}(1-p_{bi}) a_{bi}.$$

(5.14)

In form, the weighted derivative kernel of $G_t^-(\gamma)$ shares the same common factor as the logit-derivative formula of the negative class given later in Section 5.3.1; however, because the normalisation method and the treatment of numerical stability terms are different, $G_t^-(\gamma)$ is generally not identical to $\mathcal{H}_t^-(\gamma)$ defined later. Given the target negative-to-positive statistic ratio $\rho > 0$, the temporal smoothing coefficient $\beta \geq 0$, and the exponent at the previous training step $\gamma_{-,t-1}$, define the one-dimensional update objective on the current batch as

$$\Psi_t(\gamma) := \left(\log\left(G_t^-(\gamma)+\varepsilon\right) - \log\left(\rho G_t^+ + \varepsilon\right)\right)^2 + \beta(\gamma - \gamma_{-t,-1})^2, \gamma \in [\gamma_{\min}, \gamma_{\max}]. \quad (5.15)$$

During training, the constrained one-dimensional minimisation problem corresponding to $\Psi_t(\gamma)$ is taken as the criterion for determining $\gamma_{-,t}$. The variable of Ψ_t here is only the scalar γ. Accordingly, the subsequent theory is also divided into two levels: on the one hand, it analyses the comparative-static properties of the loss family $\{\ell_\gamma^-\}_{\gamma>0}$ when γ is fixed; on the other hand, it proves the continuity of the above update objective Ψ_t over the feasible interval, the existence of a global minimiser, and its degenerate form.

Thus, $\gamma_{-,t}$ is denoted as the solution to the following constrained problem:

$$\gamma_{-,t} \in \arg\min_{\gamma\in[\gamma_{\min},\gamma_{\max}]} \Psi_t(\gamma). \quad (5.16)$$

This dissertation does not further elaborate the specific numerical solution process of this constrained one-dimensional problem, but focuses only on the structure and theoretical properties of its objective function.

5.2 Objective of the Analysis

The secondary label space simultaneously exhibits three statistical characteristics: long-tail distribution, conditional rearrangement, and a high proportion of negative coordinates. The improvement of this dissertation over fixed-exponent ASL lies not in rewriting the threshold-truncation form, but in using batch-level update-statistic matching to drive a training-step-dependent negative-side focusing exponent. Section 5.1.1 gave the corresponding update objective; on this basis, the subsequent theory in this chapter continues to focus on two questions: first, what kind of comparative-static structure the negative-branch family $\{\ell_\gamma^-\}_{\gamma>0}$ has when γ changes; second, whether the objective function $\Psi_t(\gamma)$ defined by the update statistics is well-defined over the feasible interval, and into what form it degenerates when there are no active negative coordinates.

5.3 Analysis of the Fixed-Parameter Negative Branch

This section fixes a batch-level probability/logit table indexed by t and treats the negative-side focusing exponent as a constant γ. This is equivalent, within the update framework of Section 5.1.1, to performing static analysis after regarding γ as a given parameter. Accordingly, the following results discuss only the negative branch ℓ_γ^- and the statistics induced by the batch probability table.

Let the clip constant be $c \in (0,1)$ and the negative side focusing exponent be $\gamma > 0$. For any negative-label coordinate, let the predicted probability of the positive class be $p \in (0,1)$, and define

$$m(p) := (p-c)_+ \quad (5.17)$$

From

$$\tilde{p} = \min(1, 1-p+c) \quad (5.18)$$

in the training loss, one can directly verify that

$$1-\tilde{p}=(p-c)_{+}=m(p), \qquad \tilde{p}=1-m(p). \tag{5.19}$$

Therefore, the negative branch of a single negative-class coordinate can be rewritten exactly as

$$\ell_{\gamma}^{-}(p):=-m(p)^{\gamma}\log\big(1-m(p)\big) \tag{5.20}$$

Since $p<1$ and $c\in(0,1)$, we have $m(p)\in[0,1-c)\subset[0,1)$. Since this dissertation always assumes $\gamma>0$, and all asymptotic expressions below are understood under fixed γ, when $m\downarrow 0^{+}$,

$$-m^{\gamma}\log(1-m)=m^{\gamma+1}+O(m^{\gamma+2})\to 0. \tag{5.21}$$

Therefore, Equation (5.20) can be continuously extended to zero at $m=0$, and is thus always well-defined within the scope of this dissertation. In particular, when $p\le c$, we have $m(p)=0$, and therefore $\ell_{\gamma}^{-}(p)=0$; when $p>c$, Equation (5.20) is equivalent to

$$\ell_{\gamma}^{-}(p)=-(p-c)^{\gamma}\log(1-p+c) \tag{5.22}$$

Therefore, the focus of this section is to characterise the comparative-static properties of the parameterised loss family $\{\ell_{\gamma}^{-}\}_{\gamma>0}$ itself, as well as its connection with the update objective. Figure 9 gives the functional shape of the ASL-GB negative branch relative to the BCE negative branch under threshold truncation.

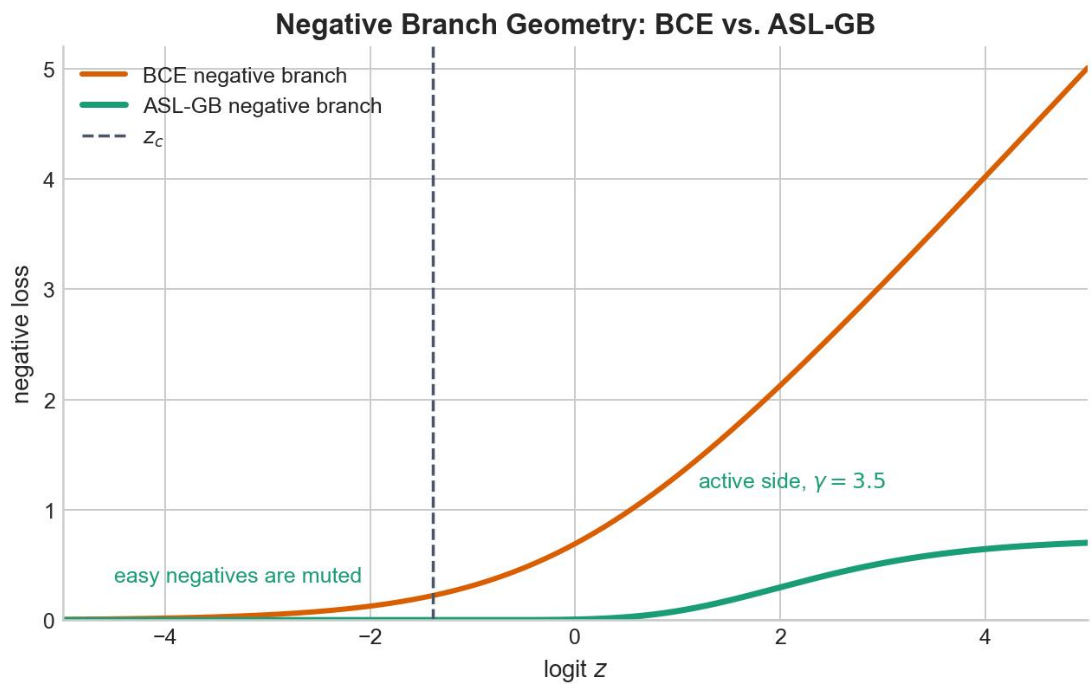

Figure 9. BCE vs. ASL-GB negative branches.

5.3.1 Notation and Batch-Level Statistics

Fix training step t, and fix one mini-batch (hereafter referred to simply as a batch). Suppose the batch contains B_{sam} samples, each with K binary labels, and let its label matrix be

$$Y = (y_{\nu,k}) \in \{0,1\}^{B_{\text{sam}} \times K}. \tag{5.23}$$

To avoid confusion with the mini-batch size B at the beginning of the chapter, the sample count is denoted specifically as B_{sam} in this subsection. Define the spaces of batch-level probability tables and logit tables paired with this label configuration as

$$\mathcal{P}(Y) := (0,1)^{B_{\text{sam}} \times K}, \qquad \mathcal{Z}(Y) := \mathbb{R}^{B_{\text{sam}} \times K}. \tag{5.24}$$

The coordinate-wise sigmoid and logit are bijections between $\mathcal{Z}(Y)$ and $\mathcal{P}(Y)$, so in what follows it suffices to construct the probability table in $\mathcal{P}(Y)$ and then take the coordinate-wise logit. For the k-th label of the ν-th sample, let the corresponding logit be

$$z_{\nu,k} \in \mathbb{R}, \tag{5.25}$$

the label be $y_{\nu,k} \in \{0,1\}$, and the corresponding predicted probability be

$$p_{\nu,k} = \sigma(z_{\nu,k}) \in (0,1). \tag{5.26}$$

To avoid overly heavy notation, fix one negative coordinate (ν,k) first, abbreviate it as (z,p), and write

$$m = (p-c)_+ \tag{5.27}$$

For any $p \in (0,1)$, let the corresponding logit be $z = \operatorname{logit}(p)$. First define, on $p \in (0,c) \cup (c,1)$, the interior derivative function of the negative branch with respect to the logit as

$$g_\gamma(p) := \frac{d}{dz} \ell_\gamma^-(\sigma(z)) \bigg|_{z=\operatorname{logit}(p)} \tag{5.28}$$

When $p < c$, $\ell_\gamma^-(p)$ is identically zero in a neighbourhood of the corresponding logit, so $g_\gamma(p) = 0$; when $p > c$, by the chain rule and $\dfrac{dp}{dz} = p(1-p)$, one obtains

$$g_\gamma(p) = \left[\gamma(p-c)^{\gamma-1}\big(-\log(1-p+c)\big) + \frac{(p-c)^\gamma}{1-p+c} \right] p(1-p) \tag{5.29}$$

To avoid repeatedly switching notation at the threshold point and the endpoints, first define the auxiliary function

$$\bar{g}_\gamma(p) := \begin{cases} 0, & p \in (0,c] \cup \{1\} \\ g_\gamma(p), & p \in (c,1) \end{cases} \tag{5.30}$$

Theorem 5.1 will prove that, if one writes $F(z) := \ell_\gamma^-(\sigma(z))$, then for any $p \in (0,1) \setminus \{c\}$ one has $F'(\operatorname{logit}(p)) = \bar{g}_\gamma(p)$, and F is differentiable at $z_c := \operatorname{logit}(c)$ with $F'(z_c) = 0$. The value at $p = 1$ is taken as a boundary extension by $\bar{g}_\gamma(1) := 0$.

The batch-level quantities to be analysed directly later are now defined. Let

$$N_t^+ := \#\{(\nu,k) : y_{\nu,k} = 1\}, \qquad N_t^- := \#\{(\nu,k) : y_{\nu,k} = 0\} \tag{5.31}$$

Here, $\#\{\cdot\}$ Denotes the number of elements in a set. A coordinate satisfying $y_{v,k} = 0$ and $p_{v,k} > c$ is called an active negative coordinate; these are exactly the negative-class coordinates whose negative branch is not removed by threshold truncation. In what follows, whenever negative-class average quantities are discussed, $N_t^- > 0$ is assumed by default. Under this condition, define the batch-averaged negative loss on negative-class coordinates as

$$\mathcal{E}_t^-(\gamma) := \frac{1}{N_t^-} \sum_{y_{v,k}=0} \ell_\gamma^-(p_{v,k}) \tag{5.32}$$

and the batch-averaged statistic of the absolute derivative as

$$\mathcal{H}_t^-(\gamma) := \frac{1}{N_t^-} \sum_{y_{v,k}=0} \bar{g}_\gamma(p_{v,k}) \tag{5.33}$$

Theorem 5.1 will prove that, for any negative-class coordinate, if $F(z) = \ell_\gamma^-(\sigma(z))$, then

$$\left| F'(\operatorname{logit}(p_{v,k})) \right| = \bar{g}_\gamma(p_{v,k}). \tag{5.34}$$

Therefore, $\mathcal{H}_t^-(\gamma)$ is exactly the average absolute value of the coordinate-wise derivative with respect to the logit for the negative class. $\mathcal{E}_t^-(\gamma)$ and $\mathcal{H}_t^-(\gamma)$ are used only to characterise the loss and derivative structure under fixed γ; the training update in Section 5.1 is still given by G_t^+, $G_t^-(\gamma)$, and $\Psi_t(\gamma)$.

It should also be noted that both $\mathcal{E}_t^-(\gamma)$ and $\mathcal{H}_t^-(\gamma)$ are batch averages of coordinate-wise nonlinear quantities, and therefore generally cannot be uniquely determined only by the mean negative-class probability

$$\bar{p}_t^- := \frac{1}{N_t^-} \sum_{y_{v,k}=0} p_{v,k} \tag{5.35}$$

Even if two batches have the same $\bar{p}_t^-$, as long as the distributions of active negative coordinates above the threshold c differ, their corresponding average negative loss and average absolute logit gradient may also differ. Figure 10 gives a simple illustration.

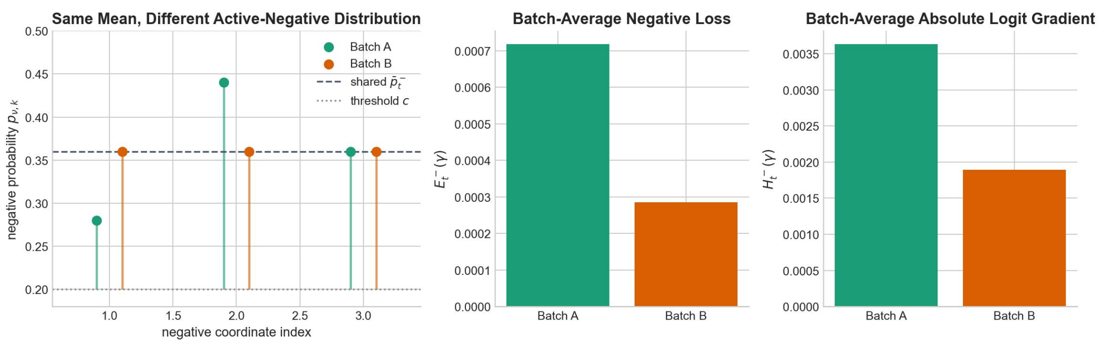

Figure 10. Negative-branch statistics under equal means.

Furthermore, to connect with the update objective in Section 5.1.1, under a fixed batch define the negative-side update statistic as

$$G_t^-(\gamma) := \frac{1}{N_t^- + \varepsilon} \sum_{y_{v,k}=0} \left[\gamma m_{v,k,\varepsilon}^{\gamma-1} \big(-\log(1-m_{v,k})\big) + \frac{m_{v,k}^{\gamma}}{1-m_{v,k}+\varepsilon} \right] p_{v,k}(1-p_{v,k}) a_{v,k}$$

(5.36)

where

$$m_{v,k} := (p_{v,k} - c)_+, \qquad a_{v,k} := \mathbf{1}\{m_{v,k} > 0\}, \qquad m_{v,k,\varepsilon} := \max\{m_{v,k}, \varepsilon\}. \tag{5.37}$$

The corresponding update objective function is

$$\Psi_t(\gamma) := \Big(\log\big(G_t^-(\gamma) + \varepsilon\big) - \log\big(\rho G_t^+ + \varepsilon\big)\Big)^2 + \beta(\gamma - \gamma_{-,t-1})^2, \gamma \in [\gamma_{\min}, \gamma_{\max}] \tag{5.38}$$

where G_t^+ is given in Section 5.1.1; under the fixed-batch condition of this section, G_t^+ is independent of γ, and can therefore be treated as a constant. It should be emphasised that Equation (5.36) contains stability terms and is generally not equal to the exact derivative average $\mathcal{H}_t^-(\gamma)$ in Equation (5.33). From here onward, only the two types of results to be proved later are retained: one concerns the exact reweighting law of the negative branch when γ changes and its effect on $\mathcal{E}_t^-(\gamma)$; the other concerns the continuity of the update objective $\Psi_t(\gamma)$ directly defined by Equation (5.38), the existence of a global minimiser, and the degenerate cases.

5.3.2 Technical Preliminaries

Theorem 5.1 (Fixed-γ Threshold and Gradient Structure)

Fix any $\gamma > 0$. Then the negative branch ℓ_γ^- and its logit derivative g_γ defined on $p \in (0,c) \cup (c,1)$ satisfy:

1. If $p < c$, then

$$\ell_\gamma^-(p) = 0, \qquad g_\gamma(p) = 0 \tag{5.39}$$

1. If $p \in (c,1)$, then ℓ_γ^- is strictly increasing with respect to p, and

$$\frac{d\ell_\gamma^-}{dp} = \gamma(p-c)^{\gamma-1}\big(-\log(1-p+c)\big) + \frac{(p-c)^\gamma}{1-p+c} > 0 \tag{5.40}$$

1. Let

$$F(z) := \ell_\gamma^-(\sigma(z)). \tag{5.41}$$

Then for any $p \in (0,1) \setminus \{c\}$,

$$\bar{g}_\gamma(p) = F'(\operatorname{logit}(p)). \tag{5.42}$$

Moreover, F is differentiable at $z_c := \operatorname{logit}(c)$ and

$$F'(z_c) = 0. \tag{5.43}$$

Hence $\bar{g}_\gamma$ is continuous on $[c,1]$, and satisfies

$$\bar{g}_\gamma(c) = 0, \qquad \bar{g}_\gamma(1) = 0 \tag{5.44}$$

Here $p=1$ is only a boundary extension point in the analysis and does not correspond to a finite logit.

4. There exists at least one $p_\gamma^\star \in (c,1)$ such that

$$\overline{g}_\gamma(p_\gamma^\star) = \max{}_{p\in[c,1]} \overline{g}_\gamma(p) \tag{5.45}$$

Proof We first prove Equation (5.39). Fix any $p<c$, and let $z_0 := \mathrm{logit}(p)$. Since σ is continuous at z_0 and $\sigma(z_0)=p<c$, there exists an open neighbourhood U of z_0 such that $\sigma(U)\subset(0,c)$. Hence for all $z\in U$, one has $m(\sigma(z))=0$, and therefore

$$\ell_\gamma^-(\sigma(z)) \equiv 0 \qquad (z\in U). \tag{5.46}$$

Thus $g_\gamma(p) = \frac{d}{dz}\ell_\gamma^-(\sigma(z))\Big|_{z=z_0} = 0$, and $\ell_\gamma^-(p)=0$, which yields Equation (5.39).

When $p\in(c,1)$, differentiating Equation (5.22) directly gives

$$\ell_\gamma^-(p) = -(p-c)^\gamma \log(1-p+c) \tag{5.47}$$

and therefore

$$\frac{d\ell_\gamma^-}{dp} = \gamma(p-c)^{\gamma-1}\big(-\log(1-p+c)\big) + \frac{(p-c)^\gamma}{1-p+c}. \tag{5.48}$$

Here $p-c>0$ and $1-p+c\in(c,1)$, so $-\log(1-p+c)>0$ and $\frac{1}{1-p+c}>0$, and hence both terms on the right-hand side are strictly positive. Therefore Equation (5.40)holds, and ℓ_γ^- is strictly increasing with respect to p.

We next prove Point 3. If $p<c$, it has already been shown in the previous step that F is identically zero in a neighbourhood of $\mathrm{logit}(p)$, and therefore

$$F'(\mathrm{logit}(p)) = 0 = \overline{g}_\gamma(p). \tag{5.49}$$

If $p\in(c,1)$, then by Equation (5.40)and the chain rule,

$$F'(\mathrm{logit}(p)) = \frac{d\ell_\gamma^-}{dp}(p)p(1-p) = g_\gamma(p) = \overline{g}_\gamma(p). \tag{5.50}$$

Therefore, for all $p\in(0,1)\setminus\{c\}$,

$$F'(\mathrm{logit}(p)) = \overline{g}_\gamma(p). \tag{5.51}$$

We now deal with the threshold point $p=c$. Let

$$z_c := \mathrm{logit}(c), \qquad F(z) := \ell_\gamma^-(\sigma(z)). \tag{5.52}$$

Since σ is strictly increasing, $z\le z_c$ if and only if $\sigma(z)\le c$, and therefore

$$F(z)=0 \qquad (z\le z_c). \tag{5.53}$$

On the other hand, under fixed γ, as $p\downarrow c$, let $m := p-c\downarrow 0^+$. From

$$-\log(1-m) = m + O(m^2), \qquad \frac{1}{1-m} = 1 + O(m), \qquad (c+m)(1-c-m) = c(1-c) + O(m)$$

(5.54)

substituted into Equation (5.29), one obtains

$$g_\gamma(c+m) = c(1-c)(\gamma+1)m^\gamma + O(m^{\gamma+1}) \to 0 \tag{5.55}$$

Moreover, since the sigmoid is differentiable at z_c and $\sigma'(z_c) = c(1-c) > 0$, when $z \downarrow z_c$ one has

$$\sigma(z) - c = c(1-c)(z-z_c) + O\big((z-z_c)^2\big). \tag{5.56}$$

Let $m(z) := \sigma(z) - c$. When $z \downarrow z_c$ and $z > z_c$, one has $m(z) = \sigma(z) - c = (\sigma(z)-c)_+ \downarrow 0$; from the above equation, $m(z) = O(z - z_c)$. In addition, from

$$-\log(1-m) = m + O(m^2) \tag{5.57}$$

it follows that

$$F(z) = \ell_\gamma^-(\sigma(z)) = m(z)^{\gamma+1} + O\big(m(z)^{\gamma+2}\big) = O\big((z-z_c)^{\gamma+1}\big). \tag{5.58}$$

Since $\gamma + 1 > 1$, it follows that

$$\frac{F(z) - F(z_c)}{z - z_c} \to 0 \qquad (z \downarrow z_c). \tag{5.59}$$

On the other hand, because $F(z) = 0$ for all $z \le z_c$, and $F(z_c) = 0$, its left derivative is also 0. Therefore, F is differentiable at z_c and $F'(z_c) = 0$. Combining this with Equation (5.55) and the fact that $g_\gamma(p) \equiv 0$ for $p < c$, it follows that $\overline{g}_\gamma$ is continuous at $p = c$.

At the right endpoint $p = 1$, the continuity of the factor inside the brackets in Equation (5.29) gives

$$\lim_{p \uparrow 1}\left[\gamma(p-c)^{\gamma-1}\big(-\log(1-p+c)\big) + \frac{(p-c)^\gamma}{1-p+c}\right] = \gamma(1-c)^{\gamma-1}(-\log c) + \frac{(1-c)^\gamma}{c},$$

(5.60)

and this limit is a finite positive number. Since $p(1-p) \to 0$, we further have

$$g_\gamma(p) \to 0 \qquad (p \uparrow 1). \tag{5.61}$$

By definition taking $\overline{g}_\gamma(1) := 0$, one obtains a continuous extension on $[c,1]$, and Equation (5.44)is proved.

Finally, Equation (5.29) shows that g_γ is continuous on $(c,1)$; combined with the boundary analysis above, $\overline{g}_\gamma$ is continuous on the closed interval $[c,1]$. Moreover, Equation (5.29) implies that for any $p \in (c,1)$ one has $\overline{g}_\gamma(p) = g_\gamma(p) > 0$, while the function values at both endpoints are zero. Therefore, by the Weierstrass theorem, $\overline{g}_\gamma$ attains its maximum on $[c,1]$, and the maximiser cannot lie at an endpoint but must lie at some interior point $p_\gamma^\star \in (c,1)$. Equation (5.45) is proved.

Figure 11 visualises the two key conclusions of Theorem 5.1 side by side: the left panel corresponds to the strictly flat region on the left side of the threshold, where $F(z)=\ell_\gamma^-(\sigma(z))\equiv 0$, and the right panel corresponds to the structure in which $\overline{g}_\gamma$ returns to zero at both $p=c$ and $p=1$, and attains an interior peak value at $p_\gamma^\star$. The curves in the figure also show that, under fixed c, increasing γ further decreases the gradient of active negative classes near the threshold; this phenomenon is provided only as an illustration, and the strict conclusions remain those of Theorem 5.1 and Theorem 5.2.

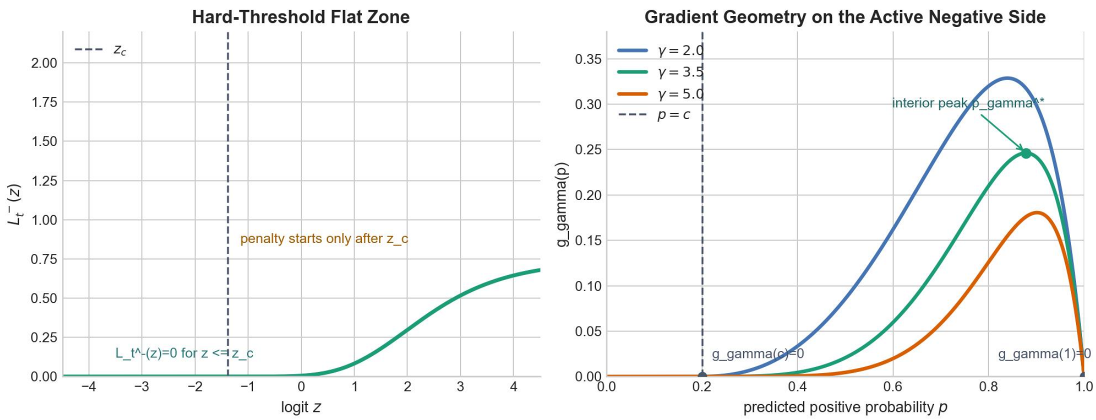

Figure 11. Theorem 5.1 illustration.

Theorem 5.1 serves only as technical preparation for what follows.

5.3.3 Comparative Statics with Respect to γ

Theorem 5.2 (Exact Negative-Branch Reweighting Law)

Let $0<\gamma_1<\gamma_2$. Then for any $p\in(c,1)$,

$$\ell_{\gamma_2}^-(p)=(p-c)^{\gamma_2-\gamma_1}\ell_{\gamma_1}^-(p)<\ell_{\gamma_1}^-(p) \tag{5.62}$$

Furthermore, for any $c<p_1<p_2<1$,

$$0<\frac{\ell_{\gamma_2}^-(p_1)}{\ell_{\gamma_1}^-(p_1)}<\frac{\ell_{\gamma_2}^-(p_2)}{\ell_{\gamma_1}^-(p_2)}<1 \tag{5.63}$$

If the current batch contains at least one active negative coordinate, namely there exists some coordinate (v,k) such that $y_{v,k}=0$ and $p_{v,k}>c$, then for any closed interval satisfying

$$I=[a,b]\subset(0,\infty),$$

with $0<a<b$, the batch-averaged negative loss $\mathcal{E}_t^-(\gamma)$ is continuous on I, differentiable on the open interval (a,b), and strictly decreasing on the whole closed interval I; moreover, for any $\gamma\in(a,b)$,

$$\frac{d}{d\gamma}\mathcal{E}_t^-(\gamma)=\frac{1}{N_t^-}\sum_{\substack{y_{v,k}=0\\ p_{v,k}>c}}\ell_\gamma^-(p_{v,k})\log(p_{v,k}-c)<0 \tag{5.64}$$

Proof For $p\in(c,1)$, rearranging Equation (5.22)directly gives

$$\ell_{\gamma_2}^-(p) = -(p-c)^{\gamma_2}\log(1-p+c) = (p-c)^{\gamma_2-\gamma_1}\ell_{\gamma_1}^-(p). \tag{5.65}$$

Since $1-p+c\in(c,1)$, one has $-\log(1-p+c)>0$, and thus $\ell_{\gamma_1}^-(p)>0$; the ratio and reweighting relation in the above equation are therefore well-defined. Since $0<p-c<1$ and $\gamma_2-\gamma_1>0$, it follows that $(p-c)^{\gamma_2-\gamma_1}\in(0,1)$, and therefore Equation (5.62) holds.

For $c<p_1<p_2<1$, the above equation still gives

$$\frac{\ell_{\gamma_2}^-(p)}{\ell_{\gamma_1}^-(p)} = (p-c)^{\gamma_2-\gamma_1}. \tag{5.66}$$

The right-hand side is a strictly increasing function of p, and therefore Equation (5.63) is obtained. This shows that increasing γ decreases the negative loss of all active negative coordinates, but suppresses more strongly those samples that are closer to the threshold c.

We next consider the batch-level negative loss. Fix any closed interval $I=[a,b]\subset(0,\infty)$ with $0<a<b$. Each term $\ell_\gamma^-(p_{v,k})$ is a continuous function of γ, and the sum contains only finitely many terms, so $\mathcal{E}_t^-(\gamma)$ is continuous on I. For any fixed active negative coordinate, the map $\gamma\mapsto\ell_\gamma^-(p_{v,k})$ is differentiable on the open interval (a,b); since the sum has finitely many terms, $\mathcal{E}_t^-(\gamma)$ is differentiable on that open interval. If $p_{v,k}\le c$, the corresponding term is identically zero; if $p_{v,k}>c$, then

$$\frac{\partial}{\partial\gamma}\ell_\gamma^-(p_{v,k}) = -(p_{v,k}-c)^\gamma\log(1-p_{v,k}+c)\log(p_{v,k}-c) = \ell_\gamma^-(p_{v,k})\log(p_{v,k}-c). \tag{5.67}$$

Since for active negative coordinates one has $\ell_\gamma^-(p_{v,k})>0$ and $0<p_{v,k}-c<1$, every active term

$$\ell_\gamma^-(p_{v,k})\log(p_{v,k}-c) \tag{5.68}$$

is strictly negative, while inactive terms are identically zero. As long as the batch contains at least one active negative coordinate, the right-hand side of Equation (5.64) is a sum of non-positive terms with at least one strictly negative term, and is therefore strictly less than zero for any $\gamma\in(a,b)$. Now take any $\gamma_a<\gamma_b$ in I. By the Lagrange mean value theorem, there exists $\xi\in(\gamma_a,\gamma_b)$ such that

$$\mathcal{E}_t^-(\gamma_b)-\mathcal{E}_t^-(\gamma_a) = \frac{d}{d\gamma}\mathcal{E}_t^-(\xi)(\gamma_b-\gamma_a)<0. \tag{5.69}$$

Hence $\mathcal{E}_t^-(\gamma)$ is strictly decreasing on the whole closed interval I. This completes the proof.

If there are no active negative coordinates in the current batch, then for all negative-class coordinates one has $p_{v,k}\le c$, and therefore $\mathcal{E}_t^-(\gamma)\equiv 0$; thus, the existence assumption in the theorem is exactly the non-degeneracy condition required for the strict monotonicity conclusion.

Figure 12 visualises the reweighting ratio in Equation (5.62)-(5.63),

$$r_{\Delta\gamma}(p) := \frac{\ell^-_{\gamma_2}(p)}{\ell^-_{\gamma_1}(p)} = (p-c)^{\Delta\gamma}, \qquad \Delta\gamma := \gamma_2 - \gamma_1 > 0, \tag{5.70}$$

as the vertical axis, taking several representative values of $\Delta\gamma$ as curve parameters. Equation (5.62)-(5.63) show that, as long as $\Delta\gamma > 0$, the corresponding reweighting ratio always lies in $(0,1)$ and increases strictly with p, so active negative coordinates closer to the threshold c are suppressed more strongly.

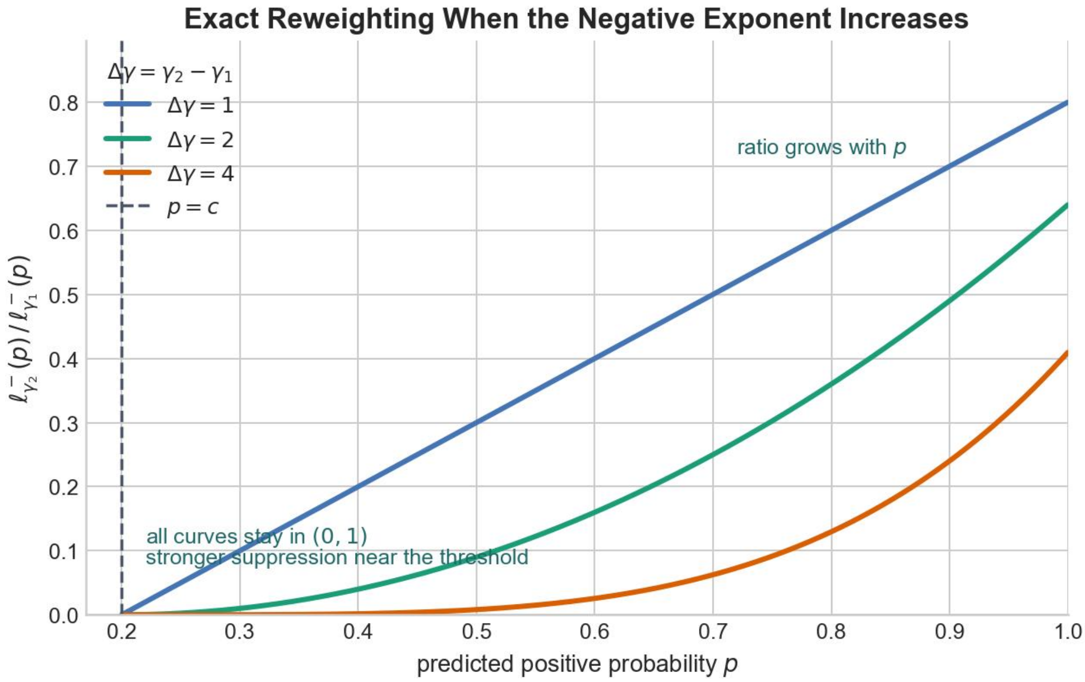

Figure 12. Reweighting ratio.

5.3.4 Existence and Degeneracy of the Update Objective

Proposition 5.3 (Existence of a Global Minimiser)

Under the notation of Section 5.1.1 and Equations (5.36)-(5.38), fix a batch satisfying $N_t^+ > 0$ and $N_t^- > 0$, and assume that the exponent at the previous training step satisfies $\gamma_{-,t-1} \in [\gamma_{\min}, \gamma_{\max}]$. Then:

1. The update objective $\Psi_t(\gamma)$ is continuous on the closed interval $[\gamma_{\min}, \gamma_{\max}]$, and therefore there exists at least one global minimiser

$$\gamma_t \in [\gamma_{\min}, \gamma_{\max}] \quad \text{such that} \quad \Psi_t(\gamma_t) = \min_{\gamma \notin \gamma_{\min}, \gamma_{\max}]} \Psi_t(\gamma) \tag{5.71}$$

2. If there are no active negative coordinates in the current batch, namely if $p_{v,k} \le c$ holds for all coordinates satisfying $y_{v,k} = 0$, then

$$G_t^-(\gamma) \equiv 0 \tag{5.72}$$

and

$$\Psi_t(\gamma) = \left(\log\varepsilon - \log(\rho G_t^+ + \varepsilon)\right)^2 + \beta(\gamma - \gamma_{-,t-1})^2 \tag{5.73}$$

Therefore, $\gamma_{-,t-1}$ itself is a feasible global minimiser; if $\beta > 0$, then it is the unique global minimiser on the feasible interval.

Proof We first prove Point 1. For each negative-class coordinate (v,k), define the single-coordinate term

$$T_{v,k}(\gamma) := (1-y_{v,k})\left[\gamma m_{v,k,\varepsilon}^{\gamma-1}\big(-\log(1-m_{v,k})\big) + \frac{m_{v,k}^{\gamma}}{1-m_{v,k}+\varepsilon}\right]p_{v,k}(1-p_{v,k})a_{v,k}. \quad (5.74)$$

Then

$$G_t^-(\gamma) = \frac{1}{N_t^- + \varepsilon}\sum_{y_{v,k}=0} T_{v,k}(\gamma). \quad (5.75)$$

Since the current batch is fixed, $p_{v,k}$, $m_{v,k}$, $a_{v,k}$, and $m_{v,k,\varepsilon}$ are all constants. If $a_{v,k} = 0$, then $T_{v,k}(\gamma) \equiv 0$, which is of course continuous. If $a_{v,k} = 1$, then $m_{v,k} > 0$, and because $0 < \varepsilon < 1$ and $m_{v,k} \in (0, 1-c)$, one has

$$m_{v,k,\varepsilon} \in [\varepsilon, 1), \qquad m_{v,k} \in (0, 1-c). \quad (5.76)$$

Therefore, the mappings

$$\gamma \mapsto m_{v,k,\varepsilon}^{\gamma-1}, \qquad \gamma \mapsto m_{v,k}^{\gamma} \quad (5.77)$$

are continuous on the interval $[\gamma_{\min}, \gamma_{\max}]$; at the same time, $1 - m_{v,k} \in (c, 1)$, so $-\log(1-m_{v,k})$ and $(1-m_{v,k}+\varepsilon)^{-1}$ are both finite constants, and thus $T_{v,k}(\gamma)$ is continuous on this interval. Since the sum contains only finitely many terms, $G_t^-(\gamma)$ is continuous on $[\gamma_{\min}, \gamma_{\max}]$. Moreover, because $G_t^-(\gamma) \ge 0$ and $\varepsilon > 0$, the function

$$\gamma \mapsto \log\big(G_t^-(\gamma) + \varepsilon\big) \quad (5.78)$$

is also continuous. Combining this with the constant term $\log(\rho G_t^+ + \varepsilon)$ and the quadratic regularisation term $\beta(\gamma - \gamma_{-,t-1})^2$ in Equation (5.38), it follows that $\Psi_t(\gamma)$ is continuous on $[\gamma_{\min}, \gamma_{\max}]$. Since this interval is compact, the Weierstrass theorem guarantees that Ψ_t attains a global minimum on it at least once, and Equation (5.71) is proved.

We next prove Point 2. If there are no active negative coordinates in the batch, then $m_{v,k} = 0$ for all negative-class coordinates, and hence $a_{v,k} = 0$. Substituting this into Equation (5.36) immediately gives $G_t^-(\gamma) \equiv 0$, namely Equation (5.72). Substituting this into Equation (5.38) gives Equation (5.73). In this case, the first term is independent of γ; if $\beta > 0$, the second term, as a strictly convex quadratic function of γ, attains its unique minimum at $\gamma = \gamma_{-,t-1}$; if $\beta = 0$, then Equation (5.73) is constant on the whole feasible interval, in which case $\gamma_{-,t-1}$ is still a feasible global minimiser. This completes the proof.

Proposition 5.3 shows that this batch-level update objective is well-defined in the non-degenerate case; when there are no active negative coordinates, the update problem degenerates into the conservative case of retaining the exponent from the previous step. As for the more strongly degenerate cases $N_t^+ = 0$ or $N_t^- = 0$, γ can likewise be kept at its previous-step value by the same line of reasoning. It should be emphasised that Proposition 5.3 proves only the continuity of the objective function itself and the existence of a global minimiser, and does not discuss the convergence or error bounds of any specific numerical solution procedure.

5.3.5 Comparison with Fixed-Exponent ASL

Compared with the fixed-exponent case, this section adds two new conclusions: first, Theorem 5.2 gives the exact pointwise scaling law of the parameterised loss family $\{\ell_\gamma^-\}_{\gamma>0}$ and proves that $\mathcal{E}_t^-(\gamma)$ is continuous, differentiable, and strictly decreasing with respect to γ; second, Proposition 5.3 shows that the objective function $\Psi_t(\gamma)$ defined by the update statistics has at least one global minimiser on the feasible interval of each non-degenerate batch, and characterises the case in which it degenerates to retaining the exponent from the previous step when there are no active negative coordinates. Therefore, the conclusions of this section are limited to the comparative-static properties of the loss under a fixed batch and the well-definedness of the update objective, and do not involve the global optimality of parameter choice or the convergence of numerical iteration. In this sense, ASL-GB remains in the objective-level imbalance-control line of work while introducing a batch-adaptive exponent update criterion [14][15][20].

To distinguish more clearly between the training-time update object and the fixed-γ analysis object in this chapter, their relationship can be summarised as follows: during training, G_t^+, $G_t^-(\gamma)$, and $\Psi_t(\gamma)$ are first constructed from the current batch, and then $\gamma_{-,t}$ is determined within the feasible interval; the theoretical part then separately analyses the negative-branch family $\{\ell_\gamma^-\}_{\gamma>0}$ under fixed γ, and proves the continuity of the same update objective Ψ_t and the existence of a global minimiser on the feasible interval.

6 Experimental Results and Discussion

This chapter answers three questions: whether the two-stage system can stably complete primary scene classification and conditional secondary label prediction; under a fixed backbone and training protocol, whether the changes in metrics mainly come from the conditional correlation structure or from the loss design; and how errors propagate through the cascade when predicted primary labels replace true primary labels. The chapter proceeds in the order of overall results, same-backbone decomposition, label-level evidence, and cascade errors, retaining only the key visualisations necessary to support the conclusions.

Before reporting the experimental results, the evaluation metrics used in Chapter 6 are defined as follows.

For primary classification, accuracy is defined as $\frac{1}{N}\sum_{i=1}^{N}1\{y_i=\hat{y}_i\}$ .For class c , precision, recall, and F1 are $P_c=\frac{TP_c}{TP_c+FP_c}$, $R_c=\frac{TP_c}{TP_c+FN_c}$, and $F1_c=\frac{2P_cR_c}{P_c+R_c}$. Macro-F1 is $\text{Macro-F1}=\frac{1}{C}\sum_{c=1}^{C}F1_c$. Weighted F1 is $\text{Weighted-F1}=\sum_{c=1}^{C}\frac{n_c}{N}F1_c$. Top-k accuracy is $\text{Top-k Accuracy}=\frac{1}{N}\sum_{i=1}^{N}1\left\{y_i\in\text{Top}_k\left(x_i\right)\right\}$. For multi-label classification, let y_{ij} and $\hat{y}_{ij}$ denote the true and predicted value of label j for sample i . Micro-F1 is $\frac{2TP}{2TP+FP+FN}$, where TP , FP , and FN are accumulated over all samples and labels. Macro-F1 is $\frac{1}{L}\sum_{j=1}^{L}F1_j$. Hamming Loss is $\frac{1}{NL}\sum_{i=1}^{N}\sum_{j=1}^{L}1\left\{y_{ij}\neq\hat{y}_{ij}\right\}$. Subset Accuracy is $\frac{1}{N}\sum_{i=1}^{N}1\{\boldsymbol{y}_i=\hat{\boldsymbol{y}}_i\}$. The label-cardinality gap is $\left|\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{L}y_{ij}-\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{L}\hat{y}_{ij}\right|$. For label j , average precision is $AP_j=\sum_{n}\left(R_{j,n}-R_{j,n-1}\right)P_{j,n}$. The mean average precision is $\text{mAP}=\frac{1}{L}\sum_{j=1}^{L}AP_j$. The oracle-cascade gap is $\text{Score}_{oracle}-\text{Score}_{cascade}$.

6.1 Overall Experimental Results

6.1.1 Results for Primary Classification

Table 11. Primary classification results.

Metric	Value
Accuracy	0.9643
Micro-F1	0.9643
Macro-F1	0.9642
Weighted F1	0.9642

Metric	Value
Top-3 Accuracy	0.9949
Top-5 Accuracy	0.9978

Accuracy and Macro-F1 are highly consistent, which is in line with the approximately balanced primary-label distribution given in Chapter 3, indicating that the errors are not concentrated in a small number of tail classes, but mainly occur between scene classes that are semantically adjacent or visually similar. The main confusion pairs are shown in Table 12.

Table 12. Main primary-class confusion pairs.

Ground-truth primary class	Predicted primary class	Count
railway	railway_station	67
railway_station	railway	55
mountain	snowberg	52
stadium	ground_track_field	48
mobile_home_park	dense_residential_area	39

Table 12 shows that first-stage errors are concentrated between semantically adjacent or spatially similar primary scenes. Such misclassifications directly change the conditioning vector received by the second stage, and manifest as cascade degradation in Section 6.4.

6.1.2 Secondary-Label Results and the Oracle-Cascade Gap

The main results of the second stage under the true-condition setting and the cascade setting are shown in Table 13. To ensure consistent comparison, the table retains only the 60-dimensional secondary-label output.

Table 13. Secondary-stage Oracle and Cascade results.

Setting	Condition source	Sample-averaged Jaccard	Micro-F1	Macro-F1	mAP	LRAP	Hamming Loss
Oracle secondary	Ground-truth primary	0.8720	0.9229	0.9295	0.9722	0.9817	0.0150
Cascade secondary	Predicted primary	0.8597	0.9132	0.9127	0.9508	0.9718	0.0169

Here, Oracle secondary corresponds to the reference upper bound of performance when the true scene condition is given, while Cascade secondary corresponds to the result when the predicted primary label is used as the condition in actual deployment. Under the Oracle setting, the model reaches Micro-F1 0.9229, Macro-F1 0.9295, mAP 0.9722, LRAP 0.9817, and Subset Accuracy 0.5219; the predicted label cardinality is 5.8950, slightly higher than the true value 5.7982, indicating that slight positive expansion still exists under the threshold 0.5.

As shown in Figure 13, Cascade exhibits consistent but limited degradation relative to Oracle. The drop in mAP is more pronounced, and the relative increase in Hamming Loss is larger, indicating that first-stage errors mainly affect ranking quality and bit-wise errors through condition switching.

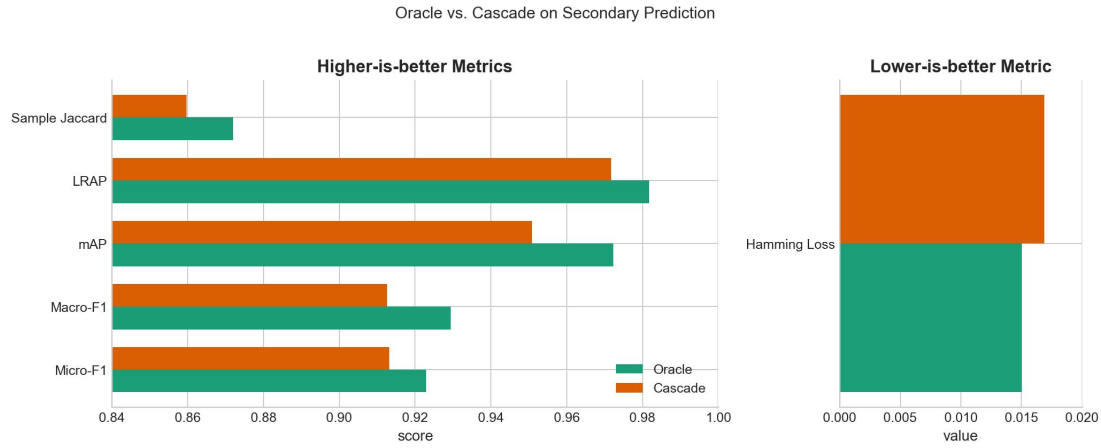

Figure 13. Oracle vs. Cascade metrics.

6.2 Sources of Performance Improvement

6.2.1 Comparison Under the Same Backbone

To separate structural factors from loss factors, this dissertation organises two layers of controlled comparison under the same test split, the same DINOv3 backbone, the same Adam (learning rate 1e-4), the same 25 epochs, and the same batch configuration. The first layer examines the setting with primary condition and a label-correlation head: structural gain is separated by comparing DINOv3 + BCE independent head with DINOv3 + BCE + conditional correlation head, and then the differences among BCE, ASL, and ASL-GB are compared under the same correlation structure. The second layer returns to the independent setting without a primary condition and without label-correlation modelling, comparing DINOv3 + BCE/ASL/ASL-GB independent head, so as to observe the independent role of the loss when no structural gain is introduced.

Structure and Loss

Table 14. Structure and loss trade-off across model settings.

Setting	Loss	Parameters / M	Micro-F1	Macro-F1	mAP	LRAP	Sample-averaged Jaccard	Subset Accuracy	Hamming Loss
DINOv3 + BCE independent head	BCE	21.62	0.9104	0.9112	0.9491	0.9656	0.8531	0.4986	0.0172
DINOv3 + BCE + conditional correlation head	BCE	29.90	0.9237	0.9295	0.9711	0.9815	0.8731	0.5249	0.0147
DINOv3 + ASL + conditional correlation head	ASL	29.90	0.9216	0.9273	0.9716	0.9816	0.8700	0.5142	0.0155
DINOv3 + ASL-GB + conditional correlation head	ASL-GB	29.90	0.9229	0.9295	0.9722	0.9817	0.8720	0.5219	0.0150

Table 14 corresponds to the first layer of comparison. Consider the structural term first: from DINOv3 + BCE independent head to DINOv3 + BCE + conditional correlation head, Micro-F1, Macro-F1, mAP, LRAP, and sample-averaged Jaccard improve by +0.0133, +0.0183, +0.0220, +0.0159, and +0.0200, respectively; Subset Accuracy improves by +0.0263, and Hamming Loss decreases by 0.00246. Therefore, the larger metric gains in the second stage come from primary condition injection and label-correlation modelling, rather than from external factors such as the number of training epochs, the optimiser, or the scale of the backbone.

Now consider the loss differences under the same correlation structure. Among BCE + conditional correlation head, ASL + conditional correlation head, and ASL-GB + conditional correlation head, the magnitude of metric change is clearly smaller than that brought by structural switching. Fixed-exponent ASL mainly pushes the model toward higher recall; after replacing it with ASL-GB, mAP, LRAP, and part of the fixed-threshold metrics are further improved, but overall it still behaves as a local reweighting adjustment on top of the existing correlation structure, rather than as a new dominant source of gain.

To further unpack the precision/recall trade-off information not listed separately in Table 14, Table 15 further lists the precision, recall, predicted label cardinality, and ranking-related metrics of the four model settings.

Table 15. Precision-recall and ranking trade-off across model settings.

Setting	Micro-Precision	Micro-Recall	Predicted label cardinality	Coverage Error	Ranking Loss	Hamming Loss
DINOv3 + BCE independent head	0.9182	0.9027	5.7008	7.0144	0.0065	0.0172
DINOv3 + BCE + conditional correlation head	0.9253	0.9222	5.7789	6.2984	0.0026	0.0147
DINOv3 + ASL + conditional correlation head	0.9044	0.9395	6.0234	6.3315	0.0027	0.0155
DINOv3 + ASL-GB + conditional correlation head	0.9153	0.9306	5.8950	6.3406	0.0028	0.0150

Table 15 further expands this trade-off. The conditional correlation head not only improves the overall metrics, but also improves precision, recall, and ranking quality simultaneously, and makes the predicted label cardinality 5.7008 closer to the true value 5.7982. Under the same correlation structure, fixed-exponent ASL raises Micro-Recall to 0.9395, but at the same time pushes the predicted label cardinality up to 6.0234; ASL-GB, while maintaining the highest mAP=0.9722 and LRAP=0.9817, pulls Micro-Precision back from 0.9044 to 0.9153, and reduces the predicted label cardinality back to 5.8950. This indicates that, relative to fixed-exponent ASL, the main role of ASL-GB is to mitigate positive expansion and rebalance precision/recall.

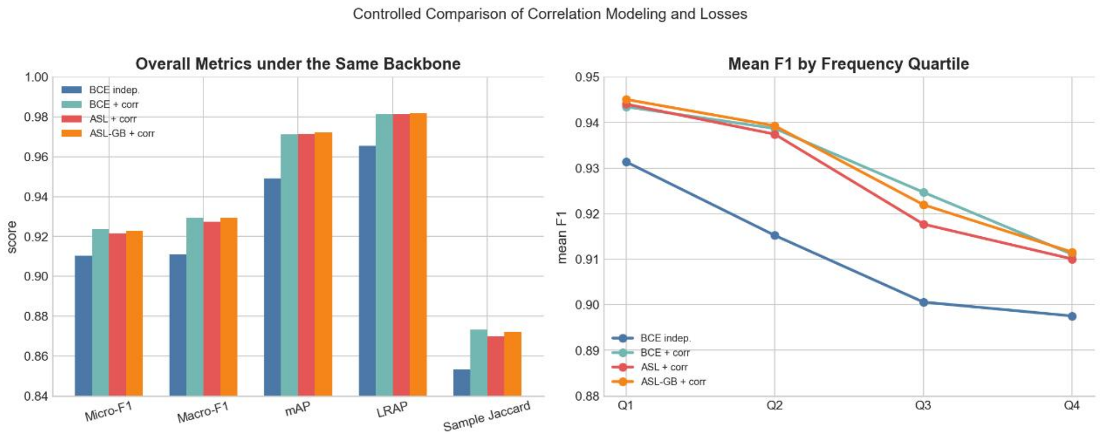

Figure 14. Structural and loss trade-off.

Independent-Head Loss Decomposition

The second layer of comparison is fixed to the independent-head setting, that is, without introducing a primary condition or label-correlation modelling. To this end, this dissertation includes DINOv3 + ASL independent head and DINOv3 + ASL-GB independent head together with DINOv3 + BCE independent head in a same-condition comparison: all three use the same data_splits, the same 25 epochs, the same AdamW (learning rate 1e-4), the same batch size (training 64 / testing 32), and all have 21.62M parameters. Therefore, the differences here can be understood more directly as the changes brought by the loss function itself.

The results are shown in the following table.

Table 16. Independent-head loss comparison.

Setting	Loss	Micro-F1	Macro-F1	mAP	LRAP	Micro-Precision	Micro-Recall	Subset Accuracy	Hamming Loss	Predicted label cardinality	Card. Gap
DINOv3 + BCE independent head	BCE	0.9104	0.9112	0.9491	0.9656	0.9182	0.9027	0.4986	0.0172	5.7008	0.0973
DINOv3 + ASL independent head	ASL	0.9133	0.9138	0.9561	0.9706	0.8925	0.9351	0.4966	0.0172	6.0752	0.2770
DINOv3 + ASL-GB independent head	ASL-GB	0.9153	0.9172	0.9562	0.9692	0.9028	0.9281	0.5020	0.0166	5.9608	0.1626

The independent-head comparison first shows that the independent gain from the loss does indeed exist. From BCE to ASL, Micro-F1, Macro-F1, mAP, and LRAP improve by +0.0029, +0.0026, +0.0069, and +0.0050, respectively, indicating that even without introducing correlation structure, asymmetric negative-side reweighting can improve detection and ranking; however, it simultaneously lowers Micro-Precision from 0.9182 to 0.8925, and pushes predicted label cardinality from 5.7008 to 6.0752, with the corresponding Card. Gap expanding from 0.0973 to 0.2770. Therefore, under the independent-head setting, fixed-exponent ASL more clearly behaves as a recall-driven trade-off.

From ASL to ASL-GB, the overall metrics continue to improve: Micro-F1 rises from 0.9133 to 0.9153, Macro-F1 rises from 0.9138 to 0.9172, Subset Accuracy rises from 0.4966 to 0.5020, and Hamming Loss falls from 0.0172 to 0.0166. More importantly, ASL-GB pulls Micro-Precision back to 0.9028, while shrinking predicted label cardinality from 6.0752 to 5.9608, reducing Card. Gap from 0.2770 to 0.1626. This shows that the training-step-dependent negative-side focusing exponent can still play a role even in an independent head without correlation structure, and its main effect remains the mitigation of the positive expansion of fixed-exponent ASL.

But the independent-head comparison also gives a boundary: although the best ASL-GB under the independent head already outperforms BCE and fixed-exponent ASL, it still remains below DINOv3 + BCE + conditional correlation head and DINOv3 + ASL-GB + conditional correlation head with correlation structure; for example, its Micro-F1=0.9153 is clearly below 0.9237 and 0.9229 of the correlation-head models, and its mAP=0.9562 is also clearly below 0.9711 and 0.9722. Therefore, loss design has independent value, but cannot replace primary condition injection and label-correlation modelling.

At the label level, relative to independent-head ASL, ASL-GB improves F1 on 44/60 labels and improves AP on 25/60 labels. Labels with more obvious gains include factory, wetland, stadium, tanks, overpass, track, park, and ships; by contrast, labels such as crosswalk, parking lot, and intersection show slight declines. Figure 15 further gives the overall metrics, error trade-offs, and label-level changes under the independent-head setting, and together with Figure 14 characterises the decomposition between structural gain and loss gain.

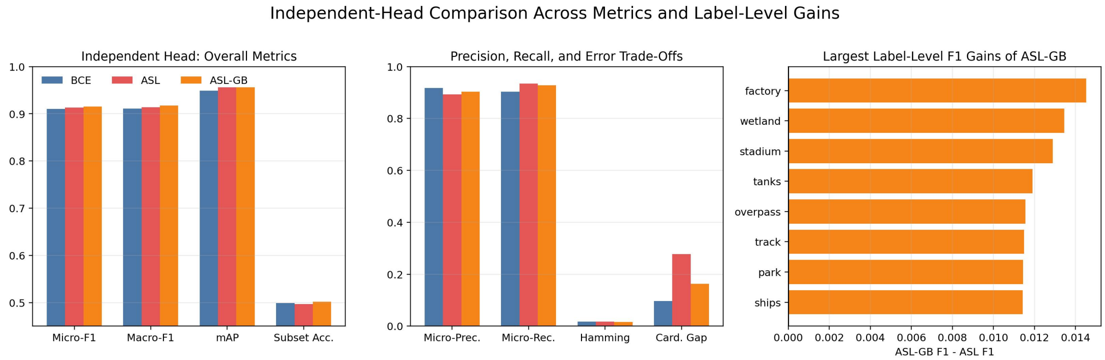

Figure 15. Independent-head loss comparison.

Combining the two layers of controlled comparison, a consistent conclusion can be reached: the major performance gains of the second stage come from primary condition injection and label-correlation modelling; the role of the loss is real, but is mainly reflected in local adjustment of ranking quality, the precision/recall trade-off, and predicted label-cardinality bias. Even under an independent head, ASL-GB outperforms fixed-exponent ASL, but the scale of its effect is still insufficient to replace structural modelling itself.

6.2.2 Performance Gains Across Frequency Groups

To examine how structural gain and loss gain are distributed across different levels of label frequency, the 60 secondary labels are divided into four prevalence quartiles according to the positive rate on the test set, producing Table 17.

Table 17. Gains by label-frequency quartile.

Frequency group	BCE F1	BCE+Corr F1	ASL+Corr F1	ASL-GB+Corr F1	BCE AP	BCE+Corr AP	ASL+Corr AP	ASL-GB+Corr AP
Q1 (lowest frequency)	0.9314	0.9435	0.9440	0.9451	0.9563	0.9696	0.9688	0.9714
Q2	0.9152	0.9387	0.9374	0.9393	0.9500	0.9769	0.9782	0.9804

Frequency group	BCE F1	BCE+Corr F1	ASL+Corr F1	ASL-GB+Corr F1	BCE AP	BCE+Corr AP	ASL+Corr AP	ASL-GB+Corr AP
Q3	0.9006	0.9247	0.9177	0.9220	0.9418	0.9722	0.9730	0.9714
Q4 (highest frequency)	0.8975	0.9111	0.9100	0.9116	0.9484	0.9659	0.9662	0.9658

From the grouping by frequency, the conditional correlation head is higher than the independent BCE head in all four frequency groups, indicating that structural gain is not confined to a small number of labels. By contrast, the effect of the loss function is smaller and more mixed: neither fixed-exponent ASL nor ASL-GB forms a monotonic advantage, but ASL-GB achieves the highest average F1 and AP in Q1 and Q2, and achieves the best overall mAP/LRAP. This further shows that, relative to the independent sigmoid head, the main improvement of the second stage comes from the structural change, while the loss function mainly adjusts the precision/recall trade-off.

6.3 Label-Level Analysis

6.3.1 Label-Wise Gains from Correlation Modelling

The conditional correlation head improves F1 on 57/60 labels and improves AP on 59/60 labels relative to the independent BCE head. The labels with the most significant improvement are shown in Table 18.

Table 18. Largest gains from the conditional correlation head.

Label	F1 increment	AP increment
factory	+0.0699	+0.0646
railway station	+0.0611	+0.0522
gully	+0.0439	+0.0327
park	+0.0431	+0.0451
parkway	+0.0427	+0.0406
overpass	+0.0404	+0.0408

Most of these labels depend on scene conditions and label-combination relationships rather than a single local texture, showing that the gain of correlation modelling is mainly concentrated on labels with strong structural dependence. This gain is not monotonic for all labels: only three classes, football field, forest, and water, do not improve in F1, among which football field has the largest F1 drop, equal to 0.0717.

6.3.2 Label-Wise Gains from ASL-GB

Under the same conditional correlation structure, after replacing fixed-exponent ASL with ASL-GB, the number of labels with improved F1 is 35/60, and the number of labels with improved AP is 32/60. This indicates that the gains of ASL-GB are not effective for all labels. Representative labels with more obvious improvement are shown in Table 19.

Table 19. Largest gains from ASL-GB.

Label	F1 increment	AP increment
bridge	+0.0232	+0.0049
ships	+0.0211	+0.0032
factory	+0.0181	+0.0118
runway	+0.0102	+0.0022
tennis court	+0.0101	+0.0029
football field	+0.0079	+0.0203

These results show that the additional gain of ASL-GB is mainly reflected in local adjustment of the precision/recall trade-off of fixed-exponent ASL, and the gains are more pronounced on labels such as factory, bridge, and football field. In contrast, labels such as crosswalk, chaparral, parking lot, and parkway show slight declines. Therefore, under the same correlation structure, ASL-GB provides a local, small-scale reweighting adjustment rather than monotonic improvement on all labels.

6.4 Analysis of Cascade Errors

The previous two sections have already shown that the predictions of the second stage depend strongly on the primary condition and the label-correlation structure. This section further discusses how this conditional dependence manifests as cascade error when the true primary label is replaced by the predicted primary label.

6.4.1 Degradation from Oracle to Cascade

Table 20 rewrites the difference between Oracle and Cascade in the form of relative change.

Table 20. Relative Oracle-to-Cascade degradation.

Metric	Oracle -> Cascade	Relative change
Micro-F1	0.9229 -> 0.9132	-1.06%
Macro-F1	0.9295 -> 0.9127	-1.81%
mAP	0.9722 -> 0.9508	-2.21%
LRAP	0.9817 -> 0.9718	-1.02%
Sample-averaged Jaccard	0.8720 -> 0.8597	-1.41%
Hamming Loss	0.0150 -> 0.0169	+12.48%

Table 20 shows that when the ground-truth primary label is replaced by the predicted primary label, the secondary-label output exhibits consistent degradation in F1, mAP, LRAP, and Jaccard, while Hamming Loss rises significantly. Combined with the first-stage accuracy of 0.9643, this decline mainly comes from switching the primary condition rather than from the overall failure of the secondary model. Cascade perturbation is more sensitive to ranking metrics, especially mAP, and to the bit-wise error rate Hamming Loss, while its effect on overall F1 is relatively smaller.

6.4.2 Labels Most Affected by Cascade Errors

After aligning the per-label evaluation results under the true-condition setting and the cascade setting, the labels with the largest F1 drops can be obtained as shown in Table 21.

Table 21. Largest Oracle-to-Cascade F1 drops.

Label	F1 under true condition	F1 under cascade condition	Difference
railway station	0.9127	0.8550	-0.0578
parkway	0.9152	0.8682	-0.0470
overpass	0.9445	0.9019	-0.0427
park	0.9460	0.9047	-0.0413
wetland	0.9060	0.8657	-0.0403
intersection	0.9854	0.9464	-0.0390
railway	0.9204	0.8819	-0.0385
factory	0.8689	0.8346	-0.0343

These labels are not necessarily all rare, but they all depend more on scene-level conditions and label-structure combinations than on a single local texture. This is consistent with the conclusions of Chapter 3 and Section 6.2, namely that the second stage performs conditional multi-label inference under the primary-scene prior.

As shown in Figure 16, the F1 drop from Oracle to Cascade is mainly concentrated on condition-sensitive labels such as railway station, parkway, and overpass, with the drop approximately in the range of 0.03 to 0.06.

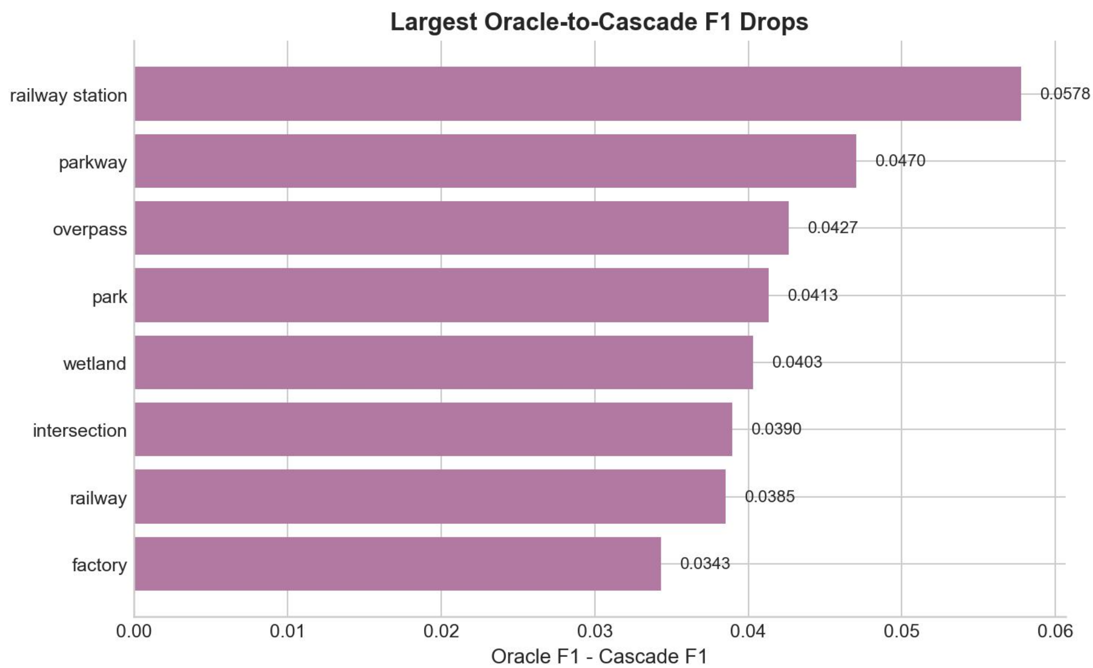

Figure 16. Largest Oracle-to-Cascade F1 drops.

Combined with the main confusion pairs of the first stage in Table 12, the two are consistent: for example, misclassifications of primary classes such as railway and railway_station, stadium and ground_track_field, and mobile_home_park and dense_residential_area directly change the conditioning vector received by the second stage. Therefore, the impact of first-stage errors should be interpreted as structured degradation caused by condition switching rather than as independent random noise.

6.4.3 Comparison with Published Results

Table 22 compares the proposed method with three published multi-label classification results on the same MLRSNet dataset [1][9][22]. Because the published studies do not report an identical metric set, the table records the best value reported in each paper's own primary metric and lists additional reported metrics where available. The comparison therefore supports a same-dataset performance positioning rather than a fully protocol-identical benchmark.

Table 22. Comparison with published multi-label classification results on MLRSNet.

Method	Backbone	Reported primary metric	Best value (%)	Additional reported values (%)	Reference
DenseNet-201	DenseNet-201	mAP	87.25	-	[1]
ResNet101-SR-Net	ResNet101 + SR-Net	CF1	87.55	CP 87.84; CR 87.26; OP 89.41; OR 87.48	[22]
S-MAT-ResNet101	ResNet101 + S-MAT	CF1	88.86	CP 88.67; CR 88.93; OP 91.21; OR 91.44	[9]
Ours (Oracle)	DINOv3 + Cond.	mAP	97.22	Micro-F1 92.29; Macro-F1 92.95; LRAP 98.17	-
Ours (Cascade)	DINOv3 + Cond.	mAP	95.08	Micro-F1 91.32; Macro-F1 91.27	-

7 Conclusion

7.1 Summary

The dissertation concludes that, for MLRSNet, modeling secondary semantic labels as a structured multi-label distribution conditioned on the primary scene label is statistically more appropriate than treating them as independent binary tasks. Empirical analysis (Chapter 3) shows that primary labels are approximately balanced, secondary labels exhibit a long-tail distribution, and the primary label significantly alters the conditional distribution of secondary labels. These observations justify the two-stage conditional decomposition proposed in Chapter 4.

Building on this, a two-stage framework based on DINOv3 is developed. The primary classification stage achieves an accuracy of 0.9643, while the secondary stage attains Micro-F1 0.9229, Macro-F1 0.9295, mAP 0.9722, and LRAP 0.9817 under oracle primary conditions. Ablation studies (Chapter 6) indicate that performance gains primarily stem from primary-condition injection and label-correlation modeling, with ASL-GB providing comparatively modest improvements. Independent-head experiments further show that ASL-GB improves Micro-F1 (0.9104 → 0.9153) and reduces the Card. Gap (0.2770 → 0.1626), yet remains inferior to models incorporating label correlations. Thus, ASL-GB has complementary value but cannot substitute structural modeling.

Mechanistically, ASL-GB adjusts the reweighting of active negative samples, primarily influencing the precision–recall trade-off under fixed-exponent ASL. Its benefits are reflected in improved ranking metrics and reduced positive over-expansion, rather than consistent gains across all evaluation metrics.

The principal limitation of the framework lies in structured cascade error. Transitioning from oracle to cascade settings results in performance degradation (Micro-F1: 0.9229 → 0.9132; mAP: 0.9722 → 0.9508; Hamming Loss: +12.48% relative). Labels with strong condition sensitivity are most affected. This highlights that while the primary label is an effective conditioning variable, hard cascade error propagation remains the dominant bottleneck.

7.2 Outcome Evaluation

Against the five objectives proposed in Chapter 1, this dissertation has completed the analysis of dataset structure, the construction of the two-stage framework, the design and analysis of ASL-GB, controlled same-backbone comparisons, and cascade error diagnosis; at the same time, it also includes ASL/ASL-GB comparisons under the independent-head baseline to further separate the contribution of the loss design itself.

In terms of research contribution, the value of this dissertation lies in aligning data statistics, model design, loss analysis, and experimental diagnosis, and thereby distinguishing the respective scopes of action of conditional correlation modelling, ASL-GB, and cascade error. The overall results show that conditional label-correlation modelling is the main source of gain, ASL-GB provides a smaller but stable trade-off adjustment; and the independent-head experiments further show that this adjustment does not rely on the correlation head to hold, but its effect size is still insufficient to replace structural modelling itself.

7.3 Limitations and Future Work

This dissertation has three main limitations: first, the current framework still adopts hard cascade, so errors in primary prediction are directly translated into structured conditioning errors; second, the main experiments concentrate on RGB images and the original annotation system of MLRSNet, and use a fixed split, so external validity still needs broader verification; third, the analysis of ASL-GB in Chapter 5 mainly stays at the batch-level loss structure and has not yet been extended to training dynamics and generalisation.

Future work can proceed in four directions: first, use soft conditioning, joint training, or explicit error correction to mitigate cascade degradation; second, combine the current correlation head with graph structure, hierarchical relationships, or label-wise adaptive thresholds to enhance modelling of condition-sensitive labels and rare labels; third, conduct more systematic label governance, repeated experiments with multiple random seeds, and cross-dataset, multimodal validation; fourth, further extend the theoretical analysis of ASL-GB to the training process, calibration, and generalisation behaviour.

7.4 Project Reflection

This project yields three main reflections. First, problem definition should precede the expansion of model complexity; if the conditional rearrangement and label dependence in MLRSNet are not identified first, subsequent method design may deviate from the task structure. Second, controlled experiments are a necessary condition for distinguishing the sources of gains; the experiments in Chapter 6 are used to separate structural modification, loss design, and cascade error. Third, boundary conditions and negative results also need to be reported; ASL-GB is not superior on all metrics, and the Oracle results also cannot directly represent deployment performance.

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Appendix A Validation on Public Benchmarks

A.1 Test-Set Statistics and Label Distributions

The statistics, tables, and visualisations in this appendix all adopt the same test-set protocol. Specifically, following the approximately iterative stratified split of 0.5 / 0.1 / 0.4 and the random seeds {13, 17, 23, 29, 31, 37, 41, 43, 47, 53}, this dissertation generates 10 test sets for each dataset; the label cardinality, label density, IRLbl, and tail-label proportion in Table 23 are all computed on these test sets and reported as mean +/- standard deviation.

Table 23 gives the statistical summary of the five public datasets on the test sets. enron has the highest mean test-set label cardinality, whereas langlog has the lowest test-set label density and the highest tail-label proportion; slashdot and enron still have high test-set max IRLbl, indicating that these benchmarks still retain clear imbalance at the test-set level.

Table 23. Test-set statistics of the five public datasets.

Dataset	Official link	Number of test-set samples	Number of labels	Input dimension	Test-set label cardinality	Test-set label density	Test-set Mean IRLbl	Test-set Max IRLbl	Test-set tail-label proportion (<5%)
birds	OpenML	259	19	271	1.0131 +/- 0.0073	0.0533 +/- 0.0004	6.25 +/- 0.55	28.70 +/- 10.59	52.11% +/- 1.66%
enron	OpenML	681	53	1001	3.3827 +/- 0.0039	0.0638 +/- 0.0001	125.41 +/- 0.59	380.00 +/- 0.47	68.49% +/- 0.91%
genbase	OpenML	265	27	1185	1.2223 +/- 0.0033	0.0453 +/- 0.0001	28.72 +/- 0.14	70.00 +/- 0.00	62.96% +/- 0.00%
langlog	OpenML	584	75	1004	1.1586 +/- 0.0062	0.0154 +/- 0.0001	29.70 +/- 0.81	68.70 +/- 0.95	89.33% +/- 0.00%
slashdot	OpenML	1513	22	1079	1.0949 +/- 0.0009	0.0498 +/- 0.0000	41.53 +/- 0.19	231.50 +/- 0.97	54.09% +/- 1.44%

Figure 17, Figure 18, Figure 19, Figure 20, and Figure 21 give the distribution of mean label prevalence over the 10 test sets. The horizontal axis is ordered from high to low according to the mean test-set label prevalence, the error bars represent the standard deviation across random seeds, and the red dashed line is the 5% tail threshold. enron and langlog have the strongest long tail on the test sets, while genbase and slashdot also retain distributions in which head labels and a large number of low-frequency labels coexist.

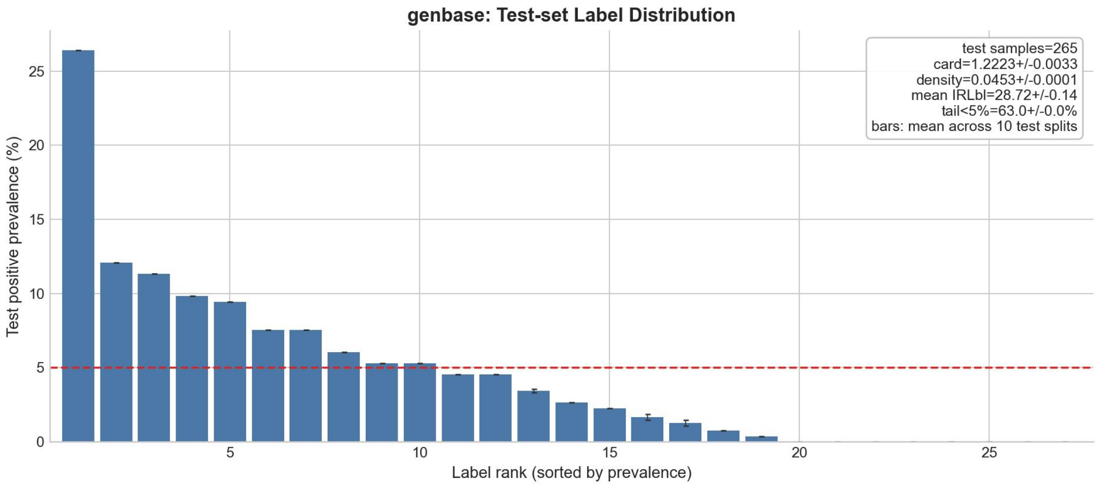

Figure 17. Genbase label prevalence.

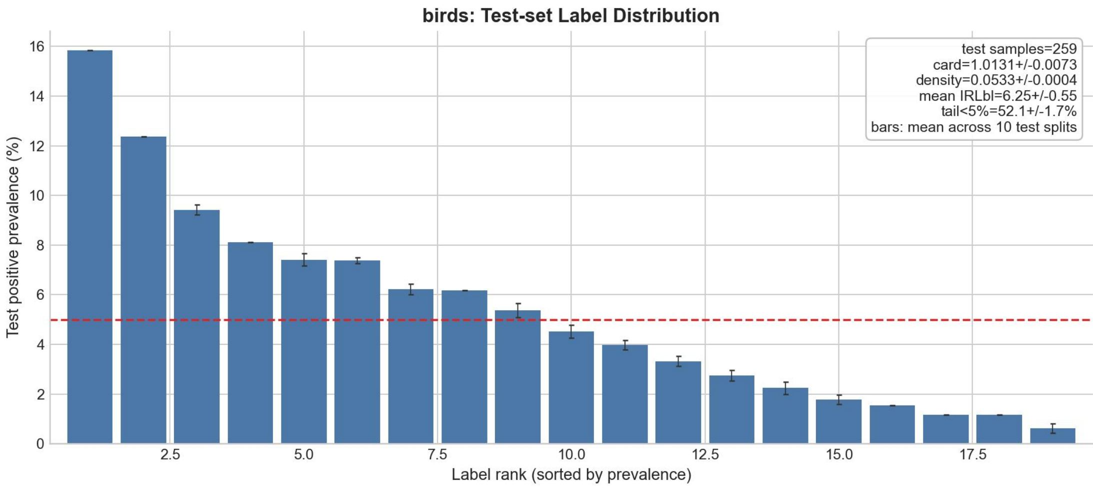

Figure 18. Birds label prevalence.

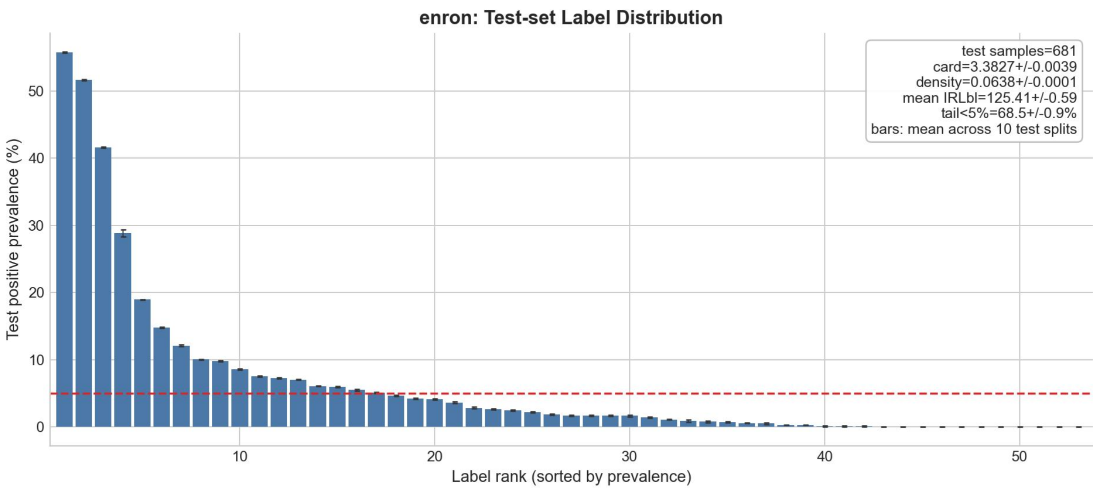

Figure 19. Enron label prevalence.

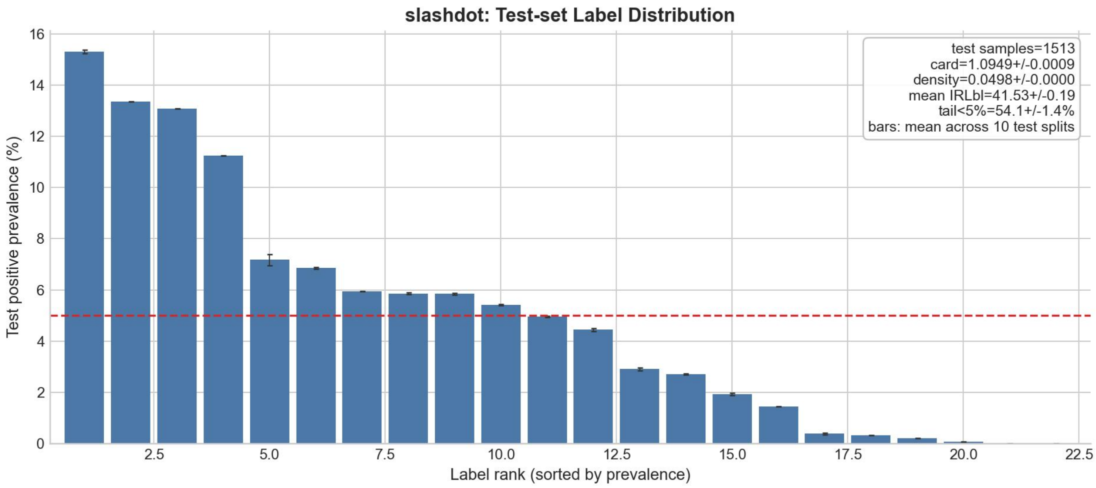

Figure 20. **Slashdot label prevalence.**

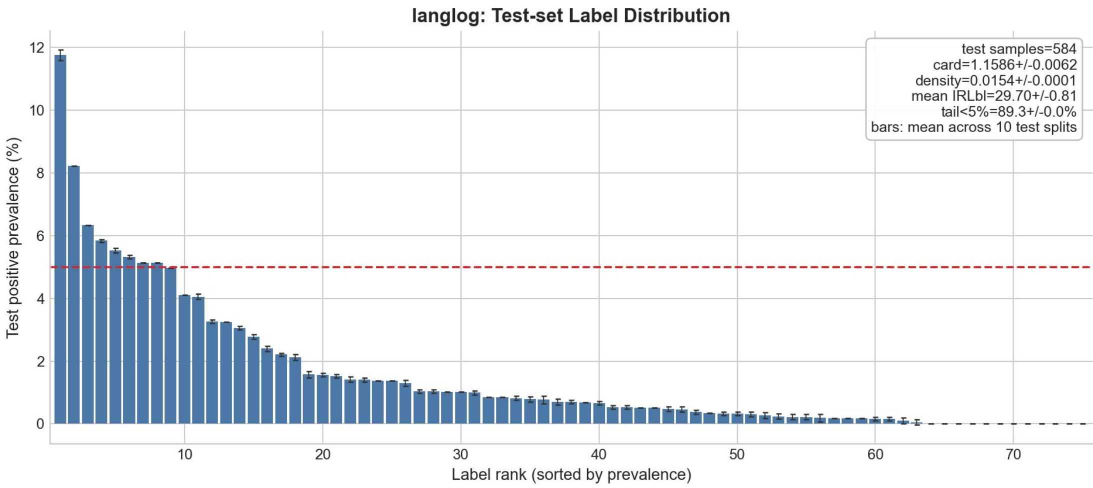

Figure 21. **Langlog label prevalence.**

A.2 Experimental Setup and Evaluation Protocol

This appendix uses the complete official datasets on OpenML, without sample deletion, class resampling, label pruning, label merging, synthetic augmentation, or manual rebalancing. In order to form a unified and reproducible controlled comparison across different loss functions, this dissertation does not reuse external fixed splits, but instead reconstructs an approximately iterative stratified train / val / test split of 0.5 / 0.1 / 0.4 for each random seed on the complete official data.

The sample set and label set remain unchanged; what changes are the input representation and the data splitting method. Specifically: (1) the input features are uniformly converted into numerical representations, with Boolean and numerical features retained directly, binary symbolic fields mapped to 0/1, and multi-valued categorical fields one-hot encoded; among the original 260 input fields of birds, 258 are already numerical columns, hasSegments is a binary field and still occupies only 1 dimension, and only the column location contains 12 categories and is expanded to 12 dimensions, so the total input dimension expands from the original OpenML 260 dimensions to 271 dimensions, while the input dimensions of the other four datasets remain consistent with the official data matrix; (2) no numerical or categorical missing values are detected in the input features of the current five public benchmarks, so there is no missing-value handling in this experiment; feature standardisation strictly fits StandardScaler only on the training set and then applies it to the validation set and the test set; (3) the label matrix directly uses the official target multi-label matrix, without deleting, merging, or renaming any labels, and only maps the original symbolic labels by column to 0/1 numerical tensors for training and evaluation.

To avoid model-structure changes interfering with the comparison of loss functions, all five datasets uniformly adopt a shared three-hidden-layer TabularMLP backbone. The network structure is

Linear -> ReLU -> Dropout(0.2) -> Linear -> ReLU -> Dropout(0.2) -> Linear -> ReLU -> Dropout(0.2) -> Linear.

The width of the first hidden layer is defined as

hidden_dim = min(1024, max(256, int(sqrt(input_dim * output_dim) * 3))),

and the second and third hidden layers are set to max(256, hidden_dim // 2) and max(128, second_dim // 2), respectively. This design ensures that under the same modelling paradigm, different datasets have capacities matched to their input and label dimensions, while avoiding separate hyperparameter tuning for any particular dataset.

Table 24 gives the instantiated network widths of the current results for each dataset.

Table 24. Instantiated network widths by dataset.

Dataset	Input dimension -> Number of labels	Hidden-layer width (h1 / h2 / h3)
genbase	1185 -> 27	536 / 268 / 134
birds	271 -> 19	256 / 256 / 128
enron	1001 -> 53	690 / 345 / 172
slashdot	1079 -> 22	462 / 256 / 128
langlog	1004 -> 75	823 / 411 / 205

The training and evaluation configuration is uniformly as follows: the train/validation/test split is 0.5 / 0.1 / 0.4; the random seeds are {13, 17, 23, 29, 31, 37, 41, 43, 47, 53}, for 10 repetitions in total; the optimiser is AdamW; the learning rate is 1e-3; the weight decay is 0; the batch size is 32; the maximum number of training steps is 250; evaluation on the validation set is performed every 10 steps, and the checkpoint with the best validation mAP-Macro is used as the final model; the test stage uses a fixed decision threshold of 0.5. Ranking metrics are computed directly from sigmoid probabilities; among them, mAP-Macro averages label AP, but if a certain label has no positive examples in the current split, that label is excluded from that average. All performance tables and the visualisations in Figure 23, Figure 24, and Figure 25 in this appendix report only test-set results; the validation set is used only for checkpoint selection.

The key settings of the three loss functions are as follows.

Table 25. Key loss-function settings.

Loss	Key configuration
BCE	Standard label-wise binary cross-entropy with logits
ASL	gamma_neg=5.0, gamma_pos=1.0, clip=0.05
ASL-GB	gamma_pos=1.0, clip=0.05, initial gamma_neg 5.0, range [3.0, 6.0], gamma_balance_ratio=0.9, gamma_reg_beta=0.15, gamma_update_lr=0.05,

Loss	Key configuration
	gamma_update_iters=2, gamma_fd_eps=0.02

To reduce the influence of single abnormal runs on the statistics, this appendix adopts the same robust rule for all test-set means and standard deviations appearing in A.3. For any dataset-loss-metric combination, let the value of the metric on the r -th test run be z_r ; first compute

$$\tilde{z} = \mathrm{median}(z), \qquad \mathrm{MAD}(z) = \mathrm{median}(| z_r - \tilde{z} |), \tag{8.1}$$

and retain non-outlier runs according to

$$| z_r - \tilde{z} | \leq 3 \times 1.4826 \times \mathrm{MAD}(z) \tag{8.2}$$

before reporting the filtered mean +/- standard deviation on the retained runs. Here

$$1.4826 = \frac{1}{\Phi^{-1}(0.75)}, \tag{8.3}$$

is the consistency correction constant that makes MAD be on the same scale as the standard deviation σ under a normal distribution, and therefore 1.4826 * MAD can be regarded as a robust estimate of the standard deviation. For Card. Gap, the calculation

$$\mathrm{Gap}_r = | \hat{c}_r - c_r | \tag{8.4}$$

is performed first on each test run, and the same rule is then applied to this sequence of absolute deviations. For the true test-set label cardinality, because the same random seed appears repeatedly under the three loss functions, this dissertation first aggregates by dataset-seed into a unique test-set cardinality before performing outlier detection. Under this rule, no outliers are detected in the current mAP-Macro, and therefore its filtered results are identical to the raw results.

A.3 Results and Visualisations

Table 26 summarises the key results of the current 10 repeated experiments on the test sets. Except for Hamming Loss and Card. Gap, the higher the value the better; Card. Gap is the absolute deviation of test-set label cardinality aggregated according to the robust rule in the previous section. The true test-set label cardinality predicted label cardinality, and all other means and standard deviations in the table are all reported as mean +/- standard deviation after outlier removal.

Table 26. Test-set results after outlier removal.

Dataset	True test-set label cardinality	Loss	Micro-F1	mAP-Macro	Subset Accuracy	Hamming Loss	Predicted label cardinality	Card. Gap
birds	1.0131 +/- 0.0073	BCE	0.4397 +/- 0.0313	0.4001 +/- 0.0325	**0.4934 +/- 0.0245**	**0.0459 +/- 0.0027**	0.5183 +/- 0.0215	0.4952 +/- 0.0183
birds	1.0131 +/- 0.0073	ASL	0.4360 +/- 0.0252	0.4049 +/- 0.0287	0.3741 +/- 0.0526	0.0835 +/- 0.0148	1.6907 +/- 0.2248	0.6787 +/- 0.2205
birds	1.0131 +/-	ASL-	**0.4901**	**0.4168**	0.4680	0.0556 +/-	1.0622 +/-	**0.0686**

Dataset	True test-set label cardinality	Loss	Micro-F1	mAP-Macro	Subset Accuracy	Hamming Loss	Predicted label cardinality	Card. Gap
	0.0073	GB	**+/- 0.0318**	**+/- 0.0414**	+/- 0.0310	0.0042	0.0665	**+/- 0.0360**
enron	3.3816 +/- 0.0023	BCE	0.5585 +/- 0.0083	0.2523 +/- 0.0182	**0.1128 +/- 0.0216**	**0.0480 +/- 0.0005**	2.3594 +/- 0.1181	1.0222 +/- 0.1187
enron	3.3816 +/- 0.0023	ASL	0.5315 +/- 0.0098	**0.2689 +/- 0.0133**	0.0467 +/- 0.0255	0.0829 +/- 0.0052	5.9865 +/- 0.4156	2.6034 +/- 0.4159
enron	3.3816 +/- 0.0023	ASL-GB	**0.5920 +/- 0.0076**	0.2554 +/- 0.0136	0.0836 +/- 0.0237	0.0548 +/- 0.0013	3.7907 +/- 0.1675	**0.4081 +/- 0.1684**
genbase	1.2223 +/- 0.0033	BCE	**0.9562 +/- 0.0214**	**0.9794 +/- 0.0206**	**0.9189 +/- 0.0395**	**0.0038 +/- 0.0018**	1.1094 +/- 0.1117	0.1145 +/- 0.1114
genbase	1.2223 +/- 0.0033	ASL	0.8295 +/- 0.1380	0.9697 +/- 0.0260	0.5106 +/- 0.3792	0.0148 +/- 0.0130	1.6043 +/- 0.3488	0.3827 +/- 0.3484
genbase	1.2223 +/- 0.0033	ASL-GB	0.9354 +/- 0.0237	0.9763 +/- 0.0211	0.8683 +/- 0.0534	0.0060 +/- 0.0023	1.3177 +/- 0.0943	**0.0901 +/- 0.0479**
langlog	1.1586 +/- 0.0062	BCE	0.1398 +/- 0.0177	0.0958 +/- 0.0068	0.1733 +/- 0.0084	**0.0153 +/- 0.0001**	0.1685 +/- 0.0257	0.9850 +/- 0.0170
langlog	1.1586 +/- 0.0062	ASL	0.2450 +/- 0.0103	**0.1282 +/- 0.0102**	0.1926 +/- 0.0018	0.0288 +/- 0.0036	1.6995 +/- 0.3530	0.5450 +/- 0.3472
langlog	1.1586 +/- 0.0062	ASL-GB	**0.2502 +/- 0.0147**	0.1226 +/- 0.0117	**0.2260 +/- 0.0133**	0.0194 +/- 0.0017	0.7812 +/- 0.1904	**0.3774 +/- 0.1907**
slashdot	1.0949 +/- 0.0009	BCE	0.4641 +/- 0.0094	0.3273 +/- 0.0129	**0.3350 +/- 0.0060**	**0.0429 +/- 0.0012**	0.6644 +/- 0.0437	0.4305 +/- 0.0435
slashdot	1.0949 +/- 0.0009	ASL	0.4336 +/- 0.0141	**0.3598 +/- 0.0099**	0.2036 +/- 0.0267	0.0850 +/- 0.0076	2.2032 +/- 0.2235	1.1083 +/- 0.2235
slashdot	1.0949 +/- 0.0009	ASL-GB	**0.4857 +/-**	0.3589 +/-	0.3155 +/-	0.0574 +/- 0.0013	1.3562 +/- 0.0426	**0.2613 +/-**

Dataset	True test-set label cardinality	Loss	Micro-F1	mAP-Macro	Subset Accuracy	Hamming Loss	Predicted label cardinality	Card. Gap
			0.0062	0.0162	0.0096			**0.0432**

Table 27 lists the outliers identified and removed according to the above rule. Here the removed random seeds and their test-set values are listed by metric-dataset-loss; no outliers are detected in mAP-Macro in the current experiment, and therefore it does not appear in Table 27.

Table 27. Identified and removed test-set outliers.

Metric	Dataset	Loss	Removed outliers
True Card.	enron	-	seed 37: 3.392071
Micro-F1	birds	ASL-GB	seed 53: 0.382488
Micro-F1	birds	BCE	seed 53: 0.277108
Micro-F1	enron	ASL	seed 31: 0.489027
Micro-F1	enron	BCE	seed 37: 0.522752
Micro-F1	genbase	ASL	seed 13: 0.474602; seed 37: 0.511591
Micro-F1	genbase	ASL-GB	seed 13: 0.870166
Micro-F1	genbase	BCE	seed 13: 0.822464; seed 17: 0.726916
Micro-F1	slashdot	ASL-GB	seed 13: 0.522602
Subset Accuracy	genbase	ASL-GB	seed 13: 0.667925
Subset Accuracy	genbase	BCE	seed 13: 0.701887; seed 17: 0.520755
Subset Accuracy	langlog	ASL	seed 13: 0.212329; seed 17: 0.169521; seed 31: 0.159247; seed 47: 0.147260
Subset Accuracy	langlog	BCE	seed 41: 0.196918
Subset Accuracy	slashdot	ASL-GB	seed 13: 0.354924
Hamming Loss	enron	ASL	seed 31: 0.104508
Hamming Loss	enron	ASL-GB	seed 31: 0.059402
Hamming Loss	enron	BCE	seed 17: 0.046325
Hamming Loss	genbase	ASL	seed 13: 0.096855; seed 37: 0.085395; seed 47: 0.069043
Hamming Loss	genbase	ASL-GB	seed 13: 0.013138
Hamming Loss	genbase	BCE	seed 13: 0.013697; seed 17: 0.019427

Metric	Dataset	Loss	Removed outliers
Hamming Loss	langlog	BCE	seed 41: 0.014932; seed 53: 0.015000
Hamming Loss	slashdot	ASL-GB	seed 13: 0.052034
Pred. Card.	birds	ASL	seed 43: 2.652510
Pred. Card.	birds	ASL-GB	seed 53: 0.664093
Pred. Card.	birds	BCE	seed 37: 0.637066; seed 53: 0.270270
Pred. Card.	enron	ASL	seed 31: 7.461087
Pred. Card.	enron	BCE	seed 37: 1.900147
Pred. Card.	genbase	ASL	seed 13: 3.750943; seed 37: 3.494340; seed 47: 3.015094
Pred. Card.	genbase	BCE	seed 17: 0.698113
Card. Gap	birds	ASL	seed 43: 1.629344
Card. Gap	birds	ASL-GB	seed 53: 0.347490
Card. Gap	birds	BCE	seed 37: 0.374517; seed 53: 0.741313
Card. Gap	enron	ASL	seed 31: 4.082232
Card. Gap	enron	BCE	seed 37: 1.491924
Card. Gap	genbase	ASL	seed 13: 2.524528; seed 37: 2.267925; seed 47: 1.796226
Card. Gap	genbase	ASL-GB	seed 13: 0.279245
Card. Gap	genbase	BCE	seed 17: 0.524528
Card. Gap	langlog	BCE	seed 53: 1.035959

Figure 23, Figure 24, and Figure 25 visualise the main information in Table 26. Figure 23 gives the per-dataset comparison of four key test-set metrics under means and standard deviations after outlier removal; Figure 24 directly compares the filtered predicted label cardinality and true label cardinality on the test set; Figure 25 gives the heat map of the filtered mean improvement of ASL-GB relative to ASL and BCE on the test set, where the column Card. Gap represents the absolute deviation on the test set after outlier removal.

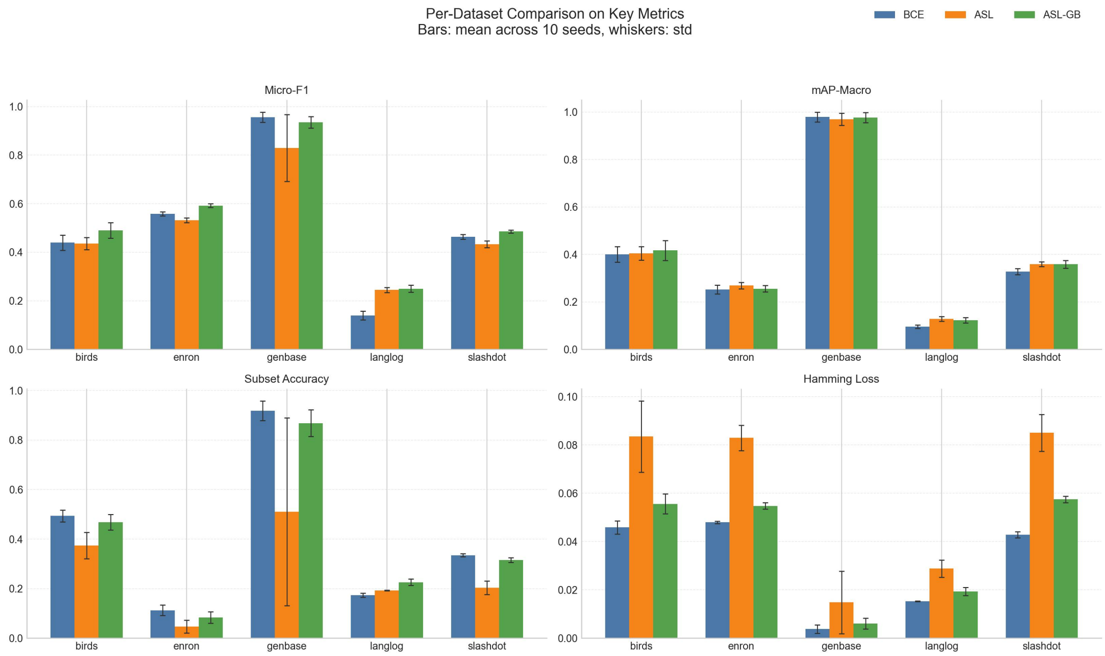

Figure 22. Key test-set metrics by dataset.

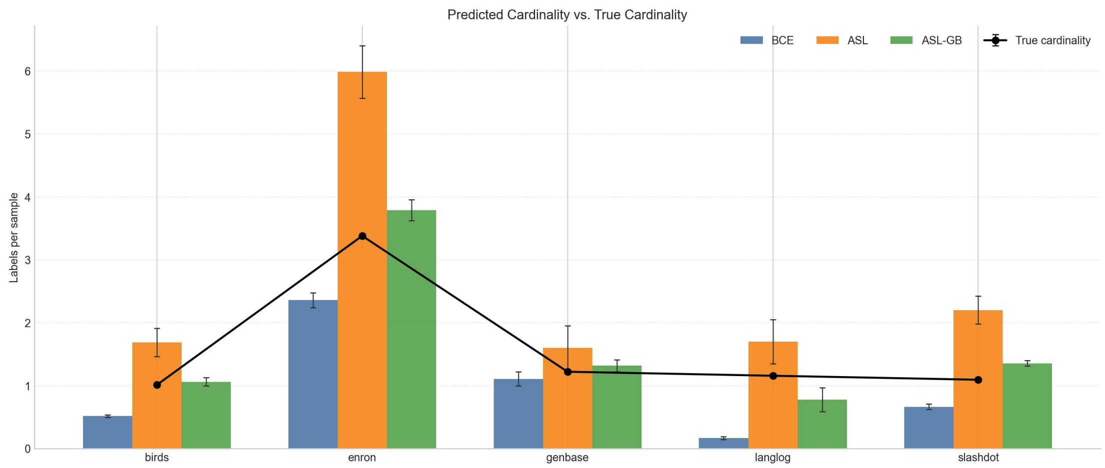

Figure 23. Predicted vs. true label cardinality.

Figure 24. ASL-GB improvement heat map.

A.4 Analysis and Conclusions

A.4.1 Changes Relative to Fixed-Exponent ASL

Comparing the filtered means after outlier removal, ASL-GB improves Micro-F1 and Subset Accuracy relative to ASL on 5/5 datasets and simultaneously improves Hamming Loss and robust Card. Gap on 5/5 datasets; mAP-Macro is higher on only 2/5 datasets. Therefore, the conclusion supported by the current test-set results is that the main role of ASL-GB relative to fixed-exponent ASL is to adjust the precision/recall trade-off, rather than monotonic improvement on all metrics.

A.4.2 Changes Relative to BCE

If the comparison target is changed to BCE, the test-set results likewise show the difference between detection gain and the cost of more conservative decisions. Comparing the filtered means after outlier removal, ASL-GB is better than BCE in Micro-F1 on 4/5 datasets, better than BCE in mAP-Macro on 4/5 datasets and achieves a smaller robust Card. Gap on 5/5 datasets. At the same time, BCE maintains lower Hamming Loss on 5/5 datasets and achieves higher Subset Accuracy on 4/5 datasets. Therefore, the main gains of ASL-GB relative to BCE are to improve detection and reduce label-cardinality bias, whereas BCE is stronger on more conservative error-control metrics.

A.4.3 Visualisation and Interpretation

The conclusions of Figure 23 and Figure 25 are consistent: under the robust filtering protocol of Section A.2, the improvements of ASL-GB are concentrated on Micro-F1, part of mAP-Macro, and Card. Gap, while Hamming Loss and most Subset Accuracy are still dominated by BCE. The current results on the five public benchmarks indicate that ASL-GB can alleviate the positive expansion of fixed-exponent ASL and reduce test-set label-cardinality bias, but cannot be regarded as a uniformly superior replacement on all metrics.